Why Should They Stop Expanding?

Computational Growth, Tipping Budgets, and the Uniformity Problem

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Abstract

Computational abundance does not by itself remove the incentive to expand. We develop a resource-allocation theory that separates physical budget growth, computational efficiency, and the creation of valued virtual worlds. With logarithmic value, concave local production, bounded external prices, and persistent residual external motives, unbounded budgets exclude permanent all-inward allocation regardless of productivity. A tight production-independent floor yields a unique tipping budget; an explicit resource-acquisition process has a unique boundary between eventual inward investment and outward lock-in. Optimizing observation and dormant backups changes the relevant condition from all-inward investment to no replication. An expected-catastrophe model shows why protecting a growing internal value can preserve a seed-backup incentive. A free-world-number lemma connects specified population ethics to value curvature, and a verified infinite-horizon benchmark separates current computational returns from future resource growth. Subjective-delay discounting can weaken remote returns, but must account for the timing of descendant production and the risk cost of delayed protection. Finally, a marked-origin model quantifies the small expansionist tail compatible with a chosen null-arrival likelihood when potential exposure is large. These results specify the resource, service-substitution, and timing tests for an inward scenario while identifying alternative routes and failures. They do not estimate extraterrestrial preferences, establish indefinite local engineering feasibility, or infer low detectability from inward allocation.

Keywords: Fermi paradox; computational inward transition; resource allocation; replication; population ethics; technosignatures.

1 Introduction

The capacity to cross interstellar distances does not settle whether a civilization will keep paying to do so. Yet an inward explanation of the Fermi paradox faces a stronger objection than this elementary distinction: resource growth itself can renew the incentive to expand. A civilization that values additional computation need not exhaust its ambitions before acquiring another star. If extensive industry produces resources and independent options that intensive local investment cannot supply cheaply, better computers may strengthen neither local sufficiency nor the decision to remain local. The computational inward transition hypothesis (CITH) therefore needs a boundary that expansion can cross, not a verbal claim that virtual experience is attractive.

Expansion arguments associated with Hart (1975) and Tipler (1980) connect technological capability to persistent settlement or self-replicating activity. The wider assumptions behind the Great Silence were surveyed by Brin (1983). Settlement dynamics, finite site lifetimes, and spatial heterogeneity complicate complete occupation (Newman and Sagan, 1981; Carroll-Nellenback et al., 2019). They do not make launch behaviour irrelevant: an expansionist lineage

can have consequences far beyond its original population. Our question is whether an internal computational alternative removes the incentive to create that lineage, and whether removing it in many civilizations is enough to explain no arrival in a specified region.

The answer developed here begins against the strong hypothesis. With logarithmic computational value, concave local production, bounded external prices, and a positive residual external motive, unbounded physical budgets eventually exclude all-inward allocation, whatever local computational productivity does. This result treats efficiency and physical resources as different quantities. A processor improvement changes the useful service obtained from a fixed energy or material allocation; access to another resource base changes the allocation that can be made. Conflating these changes would remove precisely the outward pressure that the model tests.

1.1 Contribution and organizing argument

The contribution has four connected parts. First, a common corner criterion yields a tight production-independent floor and a unique tipping budget under logarithmic value. An explicit resource-acquisition process has a unique initial-productivity boundary separating eventual inward investment from outward lock-in, with a critical path approaching the tipping budget. Second, optimizing non-replicating observation and dormant backups changes the relevant corner and replaces gross project prices by opportunity costs net of substitute services. A growing-budget construction shows that the two corners can even have opposite persistence conclusions. A catastrophe model then derives the resilience premium from value at risk rather than assigning it freely. Third, a free-world-number margin connects a specified population ethic to value curvature, while an infinite-horizon benchmark and subjective-delay inequalities make continuation returns explicit. Finally, a marked-origin arrival model quantifies how small the expansionist tail must be when potential arrival exposure is large.

These results organize a particular inward scenario around three tests: a viable resource path that stays on the appropriate side of an incentive boundary, credible non-replicating ways of satisfying external information and resilience motives, and sufficiently weak time-adjusted returns on replication. The tests must hold together in a scenario relying on all three mechanisms. They are not universal necessary conditions for every possible inward civilization. Slower marginal-value decay, vanishing external preferences, strongly increasing local returns, or rising external prices supply different routes. The model proves conditional exclusions and existence results, not a universal evolutionary destiny. Its strongest negative conclusion is correspondingly precise: within the logarithmic, concave-production benchmark, productivity alone cannot defeat a persistent residual motive when the physical budget grows without bound.

The corner calculation uses standard concave optimization. Its contribution is not a new first-order optimality theorem. The additional theory lies in sharp feasibility across production functions, the pathwise obstruction, the unique basin boundary, the exact scope and failure of an exponential endpoint test, and the links between service substitution, intertemporal returns, and population exposure. Each link has its own assumptions. In particular, a sufficient condition for one civilization to stop new investment is not a theorem that all existing autonomous descendants stop replicating. A population argument must keep their exposure in the record.

1.2 What is inherited from prior work

Computational and postbiological civilizations are established subjects, not ideas introduced by this paper (Dick, 2003; Ćirković and Bradbury, 2006). Smart (2012) describes movement toward dense internal scales. Our increasing-returns result supplies a resource-response condition against which that kind of mechanism can be tested. It does not import the stronger claim that advanced civilizations invariably leave the observable universe. Bradbury's Matrioshka-brain manuscript describes stellar computation and nested heat-use architectures (Bradbury, 1999);

the thermal calculation below is a deliberately smaller bridge between allocation and search, not a proposal of that architecture.

Aestivation instead concerns when to compute (Sandberg et al., 2017). The thermodynamic objection of Bennett et al. (2019) emphasizes that available entropy reservoirs must be specified before efficiency advantages are inferred. Our latency comparison concerns delayed external returns relative to ongoing internal activity, and does not assume that waiting for a colder universe is optimal. Sustainability and homeostatic proposals address other reasons expansion might end (Haqq-Misra and Baum, 2009; Wong and Bartlett, 2022). They can coexist with the allocation mechanism but do not prove its thresholds. Conversely, self-replicating resource acquisition (Armstrong and Sandberg, 2013) and population-total welfare (Bostrom, 2003) are direct counterpressures. They are included as cases in which outward investment can remain attractive.

1.3 Three meanings of inwardness

We reserve *all-inward allocation* for zero new expenditure on resident, industrial, and observation projects. *No replication* instead permits observation and dormant backups, while setting new reproducing capacity to zero. *Low detectability* is an observational property and can hold or fail under either allocation. These definitions also distinguish a civilization's current investment from the continued activity of earlier launches. A society may stop launching because its distant industry is already reproducing effectively; that is not an explanation of silence.

The paper proceeds through these distinctions rather than treating every external project as colonization. The model first separates physical support from economic preference, then identifies the corner and its persistence conditions. Residual motives and latency determine whether those conditions survive a more explicit account of services and time. The population calculation finally asks how much outward or leaked activity a chosen null likelihood tolerates. Detailed workload examples, the paired sensitivity ensemble, and the original two-period construction are in the supplement. Their numerical values are preserved as implementation checks, not used as a prior over unknown civilizations.

2 Model: workload, resources, and value

2.1 Experiential substitution and implemented states

Let $q \in [0,1]$ denote decision-relevant experiential fidelity and let $\sigma(q) \in [0,1]$ be an increasing substitution map. For a residence project with experiential benefit $E(n)$ and residual value $X(n)$,

$$B_{\text{phys}}(n,q) = [1-\sigma(q)]E(n) + X(n). \tag{1}$$

Perfect substitution removes the first term, not external facts, materials, geographical protection, or strategic options. For a discrete project with substitutable value e_j, residual value x_j, and incremental resource cost valued at c_j, the net comparison is

$$\Delta_j(q) = [1-\sigma(q)]e_j + x_j - c_j. \tag{2}$$

A positive reality premium need not cover its opportunity cost. The continuous model below specializes to $\sigma(q) = q$. This is a statement about a decision-maker's substitution, not about the metaphysical identity of a physical event and its simulation.

A regular hierarchy with branching factor $b \geq 2$ and finite integer depth $d \geq 0$ contains

$$W_d = \sum_{j=0}^{d} b^j = \frac{b^{d+1}-1}{b-1}. \tag{3}$$

For per-world service requirement $u_k > 0$, efficiency $\eta_k > 0$, and capacity K_k, homogeneous support requires

$$D_k(d) = u_k W_d/\eta_k \le K_k \quad \text{for every service } k. \tag{4}$$

Processing, state storage, heat rejection, and material replacement are separate constraints. Their units cannot be added without an explicit conversion. Cumulative consumption and available stocks must refer to the same horizon; instantaneous power is not a lifetime energy supply.

Lemma 1 (Finite-depth support frontier)**.** *Let $Q = \min_k K_k\eta_k/u_k$. If $Q \ge 1$, the largest feasible depth is*

$$d_{\max} = \lfloor \log_b[1 + (b-1)Q] - 1 \rfloor . \tag{5}$$

If $Q < 1$, even the root world is infeasible at the specified workload.

Proof. The intersection of the service constraints is $W_d \le Q$. Substituting the geometric sum, taking the increasing base-b logarithm, and restricting to non-negative integers gives the result. □

For fixed per-world workloads, $D_k(d+1)/D_k(d) = b + 1/W_d \to b$. With $C_d = cW_d$ and logarithmic value, the increment is $\ln\{[1 + c(bW_d + 1)]/[1 + cW_d]\} \to \ln b$. These limits divide numerator and denominator by W_d. Recursion therefore multiplies implemented demand while producing only a bounded increment in logarithmic value per additional level. A rule that generates unbounded depth is not an implementation of infinitely many simultaneous independent states. Reusing code also does not erase distinct stored histories. Scheduling fewer active worlds can save update work while leaving a growing persistent-memory bill.

On a specified horizon $[0, T]$, define

$$\chi_T = \sup_{0 \le t \le T} \max_k \frac{D_k(t)}{K_k(t)}. \tag{6}$$

Local physical sufficiency means $\chi_T \le 1$. For a ratio $\chi_{k0} \exp[(g_{D,k} - g_{K,k})t]$ initially below one and a positive growth-rate difference, its crossing time is

$$T_k = \frac{\ln(1/\chi_{k0})}{g_{D,k} - g_{K,k}}. \tag{7}$$

This follows by solving for a ratio of one. It separates a successful finite interval from a permanently feasible implementation. The archived examples and their first memory failure are retained in the supplement. The new results do not assume those examples establish indefinite support.

2.2 A common resource budget

Let $z, x, y, w \ge 0$ be local investment, resident infrastructure, industry-only capacity, and observation-only capacity. With local investment as numeraire,

$$z + p_n x + p_h y + p_m w \le B, \qquad p_j > 0, \tag{8}$$

where $B > 0$ unless the degenerate zero-budget case is expressly mentioned. A support constraint and an allocation constraint answer different questions. A physically sufficient local path can still lose to a more valuable external use, while an attractive local allocation can fail a memory or heat requirement. Their conjunction, rather than either calculation alone, describes an implementable inward path.

The planning horizon and the resource numeraire are fixed. Let B_{res} be physical resources available during that horizon and let $b_{\text{ref}} > 0$ be a fixed reference amount. The budget coordinate

is $B = B_{\text{res}}/b_{\text{ref}}$; local investment and project prices use the same resource unit. Thus $p_j x_j$ and z are resource expenditures, whereas x, y, w are dimensionless increments of project capacity. Similarly, $C = C_{\text{useful}}/c_{\text{ref}}$ uses a fixed computational-service unit $c_{\text{ref}} > 0$. In particular, logarithms take dimensionless arguments. An efficiency improvement changes the production coefficient a, not B, unless it also changes the physical resources actually available. Other physical services retain the separate constraints in equation (6).

The cumulative indices $n = 1 + x$, $h = 1 + x + y$, and $m = 1 + x + y + w$ retain the residence, industry, and observation accounting of the model. Useful computation and utility are

$$C = C_0 + a\ell(z) + G_n(x) + G_h(y), \tag{9}$$

$$U = F(C) + (1 - q)E(x) + \epsilon_R R(x + y) + \epsilon_I I(x + y + w). \tag{10}$$

Here $a > 0$, $C_0 \geq 0$, $q \in [0, 1]$, and the residual weights are non-negative. The functions E, R, I are differentiable, non-decreasing, and concave on the feasible domain. The functions G_n, G_h are continuous, non-decreasing, and concave, with $G_j(0) = 0$ and finite right derivatives $G_j'(0) = r_j \geq 0$. Set $r_m = 0$. Local production is continuous and non-negative on $[0, B]$, with $\ell(0) = 0$, and has a positive finite left derivative at B and $\ell(B) > 0$. Whenever global concavity is invoked, ℓ is also concave on this interval. These assumptions do not require a derivative at zero. The baseline is $\ell(z) = \ln(1 + z)$ and $G_j(v) = r_j v$.

We use the shifted-isoelastic computational value

$$F_\eta(C) = \begin{cases} [(1 + C)^{1-\eta} - 1]/(1 - \eta), & \eta \neq 1, \\ \ln(1 + C), & \eta = 1, \end{cases} \qquad \eta \geq 0, \tag{11}$$

so that $F_\eta'(C) = (1+C)^{-\eta}$, $F_\eta(0) = 0$, and $F_\eta'(0) = 1$. The common marginal-value normalization makes the residual weights comparable across this family. The value-curvature parameter η is distinct from the service-efficiency factors η_k. Recursive description alone does not determine a welfare functional.

The residual slopes at the all-inward corner are

$$\begin{aligned} S_n &= (1 - q)E'(0) + \epsilon_R R'(0) + \epsilon_I I'(0), \\ S_h &= \epsilon_R R'(0) + \epsilon_I I'(0), \qquad\qquad S_m = \epsilon_I I'(0). \end{aligned} \tag{12}$$

Write $d_0 = 1 + C_0 \geq 1$ and define

$$\Omega = a\ell(B), \qquad \Gamma(B) = \frac{\ell(B)}{\ell'(B)}. \tag{13}$$

The first quantity is new local computational output at the corner; the second is a resource-scale measure of production curvature. Changing the multiplicative normalization of ℓ and compensating in a changes neither quantity. Throughout, all-inward allocation means zero new expenditure in every external direction. It is stronger than zero new replication, since observation can be external without replicating.

2.3 World creation and computational-value curvature

A generative architecture does not select a welfare functional, but a free margin in the number of worlds changes the value of additional computation. To expose that margin, let $C_{\text{tot}} \geq u > 0$ be service available to worlds, let each world receive $c \geq u$, and initially relax world count to a real variable $W \geq 1$. The constraint is $Wc = C_{\text{tot}}$. The exponent $\theta \in [0, 1]$ values world population; the elasticity $\alpha \in (0, 1]$ values resources within a world. This distinction is between two margins of valuation, not between simulated and physical persons.

Lemma 2 (A free world-number margin)**.** *For the continuous relaxation,*

$$\max_{Wc=C_{\mathrm{tot}},\, W\geq 1,\, c\geq u} W^{\theta}c^{\alpha} = \begin{cases} u^{\alpha-\theta}C_{\mathrm{tot}}^{\theta}, & \theta > \alpha, \\ C_{\mathrm{tot}}^{\alpha}, & \theta \leq \alpha. \end{cases} \tag{14}$$

Thus the value exponent is $v = \max(\theta, \alpha) > 0$*. Writing* $C_{\mathrm{tot}} = u(1+C)$ *and normalizing the value at the mandatory root as* $[V(u(1+C))/V(u)-1]/v$ *gives exactly* $F_{1-v}(C)$*. For the separate specification* $W^{\theta}\ln(c/u)$*,* $\theta = 0$ *gives* $\ln(C_{\mathrm{tot}}/u)$*, while* $\theta > 0$ *gives* $(C_{\mathrm{tot}}/u)^{\theta}/(\exp(1)\theta)$ *only once* $C_{\mathrm{tot}} \geq u\exp(1/\theta)$*.*

Proof sketch. Substitution of $W = C_{\mathrm{tot}}/c$ reduces the power case to the sign of $\alpha - \theta$. In the logarithmic case, optimize $\exp[\theta\ln(C_{\mathrm{tot}}/u) - \theta v_c]v_c$ over $0 \leq v_c \leq \ln(C_{\mathrm{tot}}/u)$. The full proof, ties, and the integer-world correction are in section B.1.

At $\theta = \alpha$ every feasible continuous split ties. With integer world counts and $\theta > \alpha$, the exact value becomes $C_{\mathrm{tot}}^{\alpha}\lfloor C_{\mathrm{tot}}/u\rfloor^{\theta-\alpha}$; it has the same asymptotic exponent, not the same differentiable function. A regular recursive hierarchy restricts the available counts further but preserves this exponent. The shifted smooth family is therefore exact for the stated relaxation, not for discrete launches or indivisible worlds.

Average-log valuation is one interpretation of the baseline $\eta = 1$. It is not forced by computational sophistication. Total-log valuation with freely increasing world count instead has linear asymptotic value, and the power model with $\theta = 1$ has linear value exactly. At $\eta = 0$ the corner condition becomes

$$p_j a\ell'(B) \geq r_j + S_j. \tag{15}$$

For logarithmic local production, densification must satisfy $a/(1+B) \geq \max_j(r_j + S_j)/p_j$. More valued lives can thus strengthen the attraction of unused external resources rather than eliminate it. This connects the opportunity-cost comparison to Bostrom (2003), without attributing the present production function or equation to that paper.

Recursion supplies one implementation of an open-ended world-number margin. Its analytical role is now explicit: admitting that margin and specifying its population weight changes the induced value curvature. A non-recursive implementation with the same feasible world counts has the same implication. Recursion neither manufactures additional substrate nor chooses θ on behalf of a civilization. The extensive world margin, rather than nesting by itself, is the operative assumption.

3 The inward corner

A corner criterion is useful only if it exposes the quantities that can and cannot be changed by productivity. Temporarily allow any differentiable, strictly increasing, concave F. At the corner define

$$C_L = C_0 + a\ell(B), \qquad \lambda_L = F'(C_L)a\ell'(B), \tag{16}$$

$$D_j = F'(C_L)r_j + S_j \qquad (j = n, h, m). \tag{17}$$

Proposition 3 (The common corner criterion)**.** *Under the model assumptions, a necessary condition for the all-inward corner to be globally optimal is* $D_j \leq p_j\lambda_L$ *for every direction. This condition is also sufficient when* ℓ *is concave. For* $F = F_{\eta}$*, it is equivalent to*

$$\boxed{\frac{p_j}{\Gamma(B)} \geq \frac{r_j}{\Omega} + S_j\frac{(d_0+\Omega)^{\eta}}{\Omega}} \qquad (j = n, h, m). \tag{18}$$

Productivity enters this expression only through Ω*;* Γ *depends on the production function and the budget, not on* a*. Under concavity, strict inequalities in every direction make every optimizer all-inward. Weak inequalities establish an inward optimum but do not by themselves exclude tied external allocations.*

Proof. Let $(v_n, v_h, v_m) = (x, y, w)$. Moving a small amount $v_j > 0$ into direction j and reducing z by $p_j v_j$ gives directional derivative $D_j - p_j \lambda_L$. A positive derivative contradicts optimality. If ℓ is concave, then C is concave and composition with increasing concave F preserves concavity. Consequently,

$$\begin{aligned} U(z, x, y, w) - U(B, 0, 0, 0) &\leq \lambda_L (z - B) + \sum_j D_j v_j \\ &\leq \sum_j (D_j - p_j \lambda_L) v_j. \end{aligned} \tag{19}$$

This proves sufficiency and the strict statement. For F_η, multiplication by $(d_0 + \Omega)^\eta > 0$ gives $p_j a \ell'(B) \geq r_j + S_j (d_0 + \Omega)^\eta$; division by $\Omega > 0$ gives equation (18). □

Corollary 4 (The original all-inward criterion)**.** *For $\ell(z) = \ln(1 + z)$ and any differentiable, strictly increasing, concave F, the all-inward corner is optimal exactly when*

$$D_j / p_j \leq \lambda_L \qquad (j = n, h, m). \tag{20}$$

Proof. The local logarithm is concave, so both directions of proposition 3 apply. Its derivatives give precisely equations (16) and (17) with $\ell'(B) = 1/(1 + B)$. □

Corollary 5 (Adding hard support constraints)**.** *If an additional physical feasibility restriction retains the corner, the criterion $D_j \leq p_j \lambda_L$ remains sufficient under concavity, as does equation* (18) *for F_η. Necessity is guaranteed when each small budget-transfer direction used in the proof remains admissible. Without that condition, necessity need not hold.*

Proof. An optimum over a set is an optimum over any subset containing it. The necessity argument requires exactly the indicated feasible perturbations; without them it need not apply. □

Corollary 5.1 (The tight logarithmic floor)**.** *Let $\eta = 1$ and let ℓ be concave. Then*

$$\Gamma(B) \geq B, \tag{21}$$

with equality if and only if ℓ is linear on $[0, B]$. Every direction with $S_j > 0$ requires $p_j > S_j \Gamma(B) \geq S_j B$ at any finite a. Conversely, for fixed finite B, d_0, p_j, r_j, S_j, some concave local production function and finite $a > 0$ make the corner optimal if and only if $p_j > S_j B$ in every direction with $S_j > 0$. Linear local production provides a witness. Directions with $S_j = 0$ retain their computational-return condition but introduce no residual floor.

Proof sketch. The tangent to a concave production function at B lies above its value at zero. Equality forces coincidence with the chord on the whole interval. For the converse use $\ell(z) = z$ and take $a > \max_j (r_j + d_0 S_j)/(p_j - S_j B)$. The complete proof, including zero slopes, is in section A.1.

Corollary 6 (Logarithmic productivity thresholds)**.** *For $\eta = 1$ and concave ℓ, the exact test is*

$$p_j \geq S_j \frac{\ell(B)}{\ell'(B)} + \frac{r_j + d_0 S_j}{a \ell'(B)}. \tag{22}$$

When $p_j > S_j \Gamma(B)$ in every nontrivial direction,

$$a \geq a^\ell_{\text{crit}}(B) := \max_{j : r_j + S_j > 0} \frac{r_j + d_0 S_j}{p_j \ell'(B) - S_j \ell(B)} \tag{23}$$

is necessary and sufficient. The maximum over an empty set is zero. If a nontrivial denominator is non-positive, no finite a suffices. For $\ell(z) = \ln(1+z)$, set $L = \ln(1+B)$ and recover

$$H_j = (1+B)LS_j, \qquad A_j = (1+B)(r_j + d_0 S_j), \tag{24}$$

$$p_j \geq H_j + A_j/a, \tag{25}$$

$$a_{\text{crit}} = \max_{j:A_j>0} \frac{A_j}{p_j - H_j}. \tag{26}$$

Thus the general criterion contains both the original production test and its logarithmic threshold.

Proof. Expand the $\eta = 1$ term in equation (18) and use $\Omega = a\ell(B)$. The numerator in equation (23) is positive exactly in nontrivial directions, so rearrangement is valid exactly for positive denominators. If it is zero, then $r_j = S_j = 0$. Substitution of $\ell'(B) = 1/(1+B)$ gives the last three equations. □

The supporting-hyperplane argument is standard (Boyd and Vandenberghe, 2004, chapters 3–5). The substantive restriction is the sharp residual floor: no concave redesign of local production can reduce it below $S_j B$. Improvements in the level of computation cannot remove this obstruction at logarithmic value.

3.1 What the floor compares

The floor is a relative-value statement. An additional unit of a project can continue to provide independent information or geographical insurance while the value of more local computation declines. Raising productivity increases the level of local computation and its physical marginal product at once. Under logarithmic value these effects partly cancel: the limiting local marginal utility is $\ell'(B)/\ell(B) = 1/\Gamma(B)$, not infinity. Consequently a fixed external residual slope can remain competitive even when remote computational production becomes negligible in the comparison. This limit follows directly by dividing the logarithmic local marginal utility by a in numerator and denominator.

The distinction also explains the role of existing computation. Increasing C_0 at fixed new-investment technology does not change the asymptotic floor, but increases the finite-productivity term $d_0 S_j$ in equation (22). A civilization with more existing useful service need not place more marginal value on further service. When all residual slopes are zero, by contrast, C_0 and the choice of increasing F do not affect the computational allocation: maximizing utility is then equivalent to maximizing C. The pure-computation formula in equation (80) isolates that case. This is why an argument based only on the attractiveness of virtual experience omits an independent resource-allocation comparison.

Neither the floor nor the tipping budget measures a physical maximum number of worlds. They specify when an additional external option beats a marginal local use under the chosen objective. A system can be physically capable of supporting its present workload while economically preferring more territory. It can also prefer inward investment that fails an independent support requirement. In an application the service ratios and the allocation inequalities must therefore be evaluated on the same resource history, rather than selecting a favourable budget for one and a favourable efficiency for the other. The joint archived example demonstrates compatibility on a finite interval; the persistence results determine what additional conditions are needed beyond it.

A dimensionless robustness diagnostic is

$$\rho = \max_j \frac{H_j + A_j/a}{p_j}. \tag{27}$$

The corner is inward when $\rho \leq 1$. If every external price is multiplied by $\kappa > 0$ while other terms are held fixed, the corner survives exactly when $\kappa \geq \rho$. Thus reporting ρ identifies how

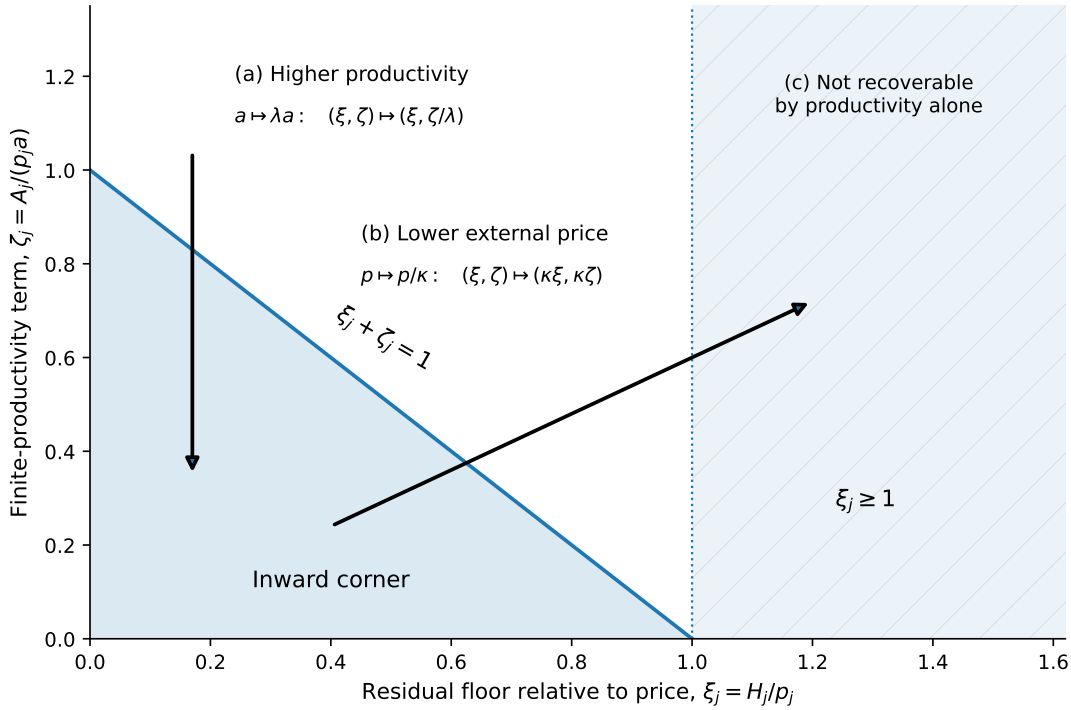


Figure 3: Exact logarithmic phase diagram and comparative statics for one external direction. The shaded triangle satisfies $H_j/p_j + A_j/(p_j a) \leq 1$; every external direction must satisfy this condition for the all-inward corner to be optimal. (a) Higher productivity reduces only ζ_j, producing a downward vertical move. (b) A price reduction to p_j/κ multiplies both coordinates by κ, producing a radial move away from the origin. (c) In the hatched region $\xi_j \geq 1$, no finite productivity restores the corner. For $A_j > 0$, $(1, 0)$ is only a limiting point. The arrows are schematic evaluations of the exact transformations; no archived inputs are overlaid and no population frequency is estimated.

much a conclusion depends on proximity to a threshold. This is conditional price sensitivity, not a distribution over unknown technologies.

3.2 Analytical phase boundary

For logarithmic value, define

$$\xi_j = \frac{H_j}{p_j}, \qquad \zeta_j = \frac{A_j}{p_j a}. \tag{28}$$

The exact phase boundary is $\xi_j + \zeta_j = 1$. All-inward allocation requires $\xi_j + \zeta_j \leq 1$ for *every* direction. If $\xi_j \geq 1$, no finite productivity restores the corner, because productivity lowers only ζ_j. Here $\xi_j \geq 1$ entails $S_j > 0$ and hence $A_j > 0$, so ζ_j remains strictly positive at every finite a. Increasing a to λa, with $\lambda > 1$, moves (ξ_j, ζ_j) vertically to $(\xi_j, \zeta_j/\lambda)$. Reducing the external price to p_j/κ, with $\kappa > 1$ and other parameters fixed, moves it radially to $(\kappa\xi_j, \kappa\zeta_j)$. The degenerate case $A_j = 0$ has $H_j = 0$ and imposes no restriction. Figure 3 shows this analytical geometry rather than an ensemble frequency.

3.3 Shared technological change

Corollary 7 (Shared gains and the phase coordinates)**.** *Fix B, d_0, S_j and logarithmic local production and value. Under $(a, r_j, p_j) \mapsto (\lambda_a a, \lambda_r r_j, p_j/\kappa)$, with $\lambda_a, \lambda_r, \kappa > 0$, equations* (24) *and* (28) *give*

$$\xi_j' = \kappa \xi_j, \qquad \zeta_j' = \frac{\kappa}{\lambda_a}\left[\zeta_j + \frac{(1+B)(\lambda_r - 1)r_j}{p_j a}\right]. \tag{29}$$

The identity also covers zero slopes; see section A.11. Cheaper external projects enlarge the productivity-independent floor. Productivity cannot offset this increase in ξ_j. Manufacturing improvements affecting both local substrates and replicating platforms therefore do not guarantee inwardness; no particular implementation is assumed. The archived Table S2 illustrates the distinction: local productivity multiplied by ten changes 219 inward cases to 314; shared productivity gives 222; the external price-floor factor 0.01 gives zero.

4 Persistence: growing versus bounded budgets

4.1 Unbounded physical budgets

Proposition 8 (Growth obstruction and sharp productivity scaling)**.** *Let $B(t) \to \infty$, let ℓ be concave, and retain the assumptions of proposition 3. Suppose that eventually some direction has $p_j(t) \le M < \infty$ and $S_j(t) \ge s > 0$. For $\eta \ge 1$, the all-inward corner is excluded at every sufficiently late time, for every path of finite positive productivities $a(t)$. For $0 \le \eta < 1$, eventual inwardness instead requires*

$$\Omega(t) \ge \left[\frac{s\Gamma(B(t))}{M}\right]^{1/(1-\eta)} \ge \left[\frac{sB(t)}{M}\right]^{1/(1-\eta)}. \tag{30}$$

For the sufficiency statement, additionally fix all $p_j > 0$, finite $r_j, S_j \ge 0$, and d_0. If $0 < \eta < 1$, it is sufficient that, eventually,

$$\Omega(t) \ge c_\Omega \Gamma(B(t))^{1/(1-\eta)}, \qquad c_\Omega > \left(\max_j \frac{S_j}{p_j}\right)^{1/(1-\eta)}. \tag{31}$$

The inequalities then become strict. For $\eta = 0$, the exact condition is

$$\frac{\Omega(t)}{\Gamma(B(t))} = a(t)\ell'(B(t)) \ge \max_j \frac{r_j + S_j}{p_j}. \tag{32}$$

In particular, for $\ell(z) = z$ and $0 < \eta < 1$, $a(t) \ge c_\Omega B(t)^{\eta/(1-\eta)}$ with the strict coefficient condition above is eventually sufficient.

Proof sketch. For $\eta = 1$ the residual term has positive infimum one after division by $S_j\Gamma$. For $\eta > 1$ that infimum is

$$\min_{\Omega > 0} \frac{(d_0 + \Omega)^\eta}{\Omega} = d_0^{\eta-1} \frac{\eta^\eta}{(\eta-1)^{\eta-1}}. \tag{33}$$

Both conflict with $\Gamma(B) \ge B \to \infty$. Below unit curvature, comparison with $\Omega^{\eta-1}$ gives the necessary rate; the strict coefficient in equation (31) leaves an asymptotic margin for remote computational returns. See section A.2.

Remark (Bounded productivity on growing budgets). Under proposition 8, $a \le a_{\max} < \infty$ gives $\Omega = O(B)$ by concavity. For $0 < \eta < 1$, this contradicts equation (30); an eventually inward path requires $a(t) \to \infty$. At $\eta = 0$, equation (32) still uses actual productivity; its necessary ceiling test is $a_{\max}\ell'(B(t)) \ge \max_j (r_j + S_j)/p_j$. A positive right-hand side and vanishing local marginal production therefore exclude persistence. See section A.9.

This result concerns the growth of physical resources, not the growth of computational efficiency. With persistent residual value and bounded prices, productivity can compensate for an unbounded resource budget only on the $\eta < 1$ branch of the concave-production model. The endpoint $\eta = 0$ must retain r_j as well as S_j; remote computation is not a vanishing correction in that case.

Corollary 9 (Logarithmic pathwise specialization)**.** *Let $\ell(z) = \ln(1+z)$ and $\eta = 1$, with fixed r_j, d_0 and finite positive $B(t), a(t), p_j(t)$. Define $H_j(t), A_j(t)$ by equation (24). On any specified path, the corner is optimal at every time exactly when, in every direction with $A_j(t) > 0$,*

$$p_j(t) > H_j(t), \qquad a(t) \ge \max_{j:A_j(t)>0} \frac{A_j(t)}{p_j(t) - H_j(t)}. \tag{34}$$

In particular, where $S_j(t) > 0$,

$$\frac{p_j(t)}{S_j(t)} > [1 + B(t)] \ln[1 + B(t)]. \tag{35}$$

For $B(t) \to \infty$, fixed positive residual slopes therefore require prices at least of order $B \ln B$; bounded prices require $S_j(t) = O((B \ln B)^{-1})$. These rate restrictions do not replace the productivity inequality. A zero-slope direction with $r_j > 0$ requires $a(t) \ge [1 + B(t)]r_j/p_j(t)$.

Proof. Apply equations (25) and (26) at each time. Non-negativity and $d_0 > 0$ imply $A_j = 0$ exactly when $r_j = S_j = 0$. Division by a positive slope and $(1 + B)\ln(1 + B) \sim B \ln B$ give the rate statements. □

4.2 What increasing returns would have to accomplish

Proposition 10 (At least exponential local returns)**.** *Let $\eta = 1$ and retain the model assumptions apart from local concavity. Let a fixed local production function satisfy $\ell(0) = 0$ and be continuously differentiable and positive with positive derivative for positive budgets. Let $B(t)$ be continuous with $B(t) \to \infty$. If eventually $p_j(t) \le M$, $S_j(t) \ge s > 0$, and the corner is optimal at every time, then for all sufficiently large B,*

$$\frac{\mathrm{d} \ln \ell(B)}{\mathrm{d}B} > \frac{s}{M}, \qquad \ell(B) > \ell(B_0) \exp\left[\frac{s}{M}(B - B_0)\right] \quad (B > B_0). \tag{36}$$

Convex local production gives $\Gamma(B) \le B$, but polynomial superlinearity is insufficient: $\ell(z) = z^k$ gives $\Gamma = B/k$. By contrast, $\ell(z) = [\exp(\psi z) - 1]/\psi$, $\psi > 0$, gives $\Gamma = [1 - \exp(-\psi B)]/\psi < 1/\psi$.

Proof sketch. The necessary strict floor gives $\ell'/\ell > S_j/p_j \ge s/M$. Continuity of the budget path covers every sufficiently large budget, so integration proves the exponential bound. The remaining identities follow by differentiation; see section A.3.

The required increase is with local resource investment, not merely with calendar time. This supplies a quantitative test for compression-based inward development proposals such as Smart (2012). Increasing returns remove the concavity guarantee. For exponential local production with one affine external channel and linear residual value, an exact global test is available: the corner must satisfy its first-order condition and outperform the all-external endpoint (proposition 29). Smooth concave residual utilities, or smooth concave external production, can invalidate this two-test shortcut. Explicit counterexamples are given in section A.4.

4.3 The tipping budget

Proposition 11 (A unique productivity-independent tipping budget)**.** *Let $\eta = 1$ and $\ell \in C^2([0,\infty))$, with $\ell(0) = 0$, $\ell' > 0$, and $\ell'' \le 0$. Fix $d_0 \ge 1$, all $p_j > 0$, and finite $r_j, S_j \ge 0$. If at least one $S_j > 0$, define B_{tip} by*

$$\Gamma(B_{\text{tip}}) = \min_{j:S_j>0} \frac{p_j}{S_j}. \tag{37}$$

This budget exists and is unique, since

$$\Gamma(0) = 0, \qquad \Gamma'(B) = 1 - \frac{\ell(B)\ell''(B)}{\ell'(B)^2} \ge 1. \tag{38}$$

The corner is attainable at finite productivity exactly for $0 < B < B_{\text{tip}}$, where its threshold is $a^{\ell}_{\text{crit}}(B)$ from equation (23). This threshold is non-decreasing and diverges as $B \uparrow B_{\text{tip}}$. At $B = B_{\text{tip}}$ as well as above it, no finite productivity makes the corner optimal. If eventually $B(t) \le B' < B_{\text{tip}}$ and $a(t) > a^{\ell}_{\text{crit}}(B')$, every static optimizer is eventually all-inward. If every $S_j = 0$, set $B_{\text{tip}} = \infty$; no finite productivity-independent residual barrier exists.

Proof sketch. Differentiate Γ. Below its unique crossing, every denominator in equation (23) is positive. Each nontrivial threshold is non-decreasing because its denominator has derivative $p_j\ell'' - S_j\ell' \le 0$. A binding positive-slope denominator vanishes at the crossing. See section A.5.

Corollary 12 (The effective budget under a productivity ceiling)**.** *Retain proposition 11 with some $S_j > 0$ and fix $0 < a_{\max} < \infty$. Define*

$$B^*(a_{\max}) = \sup\{B \in (0, B_{\text{tip}}) : a^{\ell}_{\text{crit}}(B) \le a_{\max}\}, \qquad \sup \varnothing := 0. \tag{39}$$

An inward optimum at some $0 < a \le a_{\max}$ exists exactly for $0 < B \le B^*(a_{\max})$. If $B^* > 0$, its endpoint is included and $a^{\ell}_{\text{crit}}(B^*) = a_{\max}$; an actual productivity must still meet equation (23). Moreover, $B^* < B_{\text{tip}}$, B^* is non-decreasing in $a_{\max}$, and $B^* \to B_{\text{tip}}$ as $a_{\max} \to \infty$.

An efficiency redesign can raise this effective budget, but never beyond $B_{\text{tip}} \le \min_{j:S_j>0} p_j/S_j$, since $\Gamma(B) \ge B$. The empty-set and equality cases are proved in section A.8.

Corollary 13 (Fixed- and bounded-budget curvature)**.** *At a fixed positive budget and under concave production, each nontrivial direction has a unique finite high-productivity threshold for $0 \le \eta < 1$. At $\eta = 1$, equation (23) applies. At $\eta > 1$, a direction with $S_j > 0$ has an inward set in $a > 0$ that is empty, a singleton, or a bounded closed interval. A direction with $S_j = 0$ requires only $a \ge r_j/[p_j\ell'(B)]$ for every η. For the local logarithm the test is precisely*

$$r_j + S_j(d_0 + aL)^{\eta} \le \frac{p_j a}{1+B}, \qquad L = \ln(1+B). \tag{40}$$

For the path statements fix the parameters and assume ℓ is C^1 with positive derivative near $B(t) \to \bar{B} \in (0,\infty)$. For $\eta < 1$, eventually exceeding a sufficiently large finite productivity threshold gives eventual strict inwardness. This also holds for any $0 < B(t) \le B_{\max} < \infty$ under concavity and $\ell'(B_{\max}) > 0$. For $\eta > 1$ and some $S_j > 0$, eventually exceeding a sufficiently large finite threshold instead excludes inwardness near the positive limit. At $\eta = 1$, $\bar{B} < B' < B_{\text{tip}}$ and eventual $a(t) > a^{\ell}_{\text{crit}}(B')$ suffice; $\bar{B} > B_{\text{tip}}$ excludes inwardness. At equality the pointwise comparison with $a^{\ell}_{\text{crit}}(B(t))$ remains indispensable.

Proof. See section A.6. This is the fixed-budget curvature classification, with a uniform bound near the positive limiting budget; it is not a new assumption about preferences. "No tipping budget" for $\eta < 1$ means no productivity-independent finite obstruction, not absence of a budget threshold at a fixed productivity.

For $\eta > 1$, computational value saturates while residual motives persist: sufficiently large efficiency gains reduce the marginal value of local investment and eventually favour an external deviation, not permanent inwardness.

4.4 Resource acquisition and the inward basin

Separate resources available locally from those supplied by prior replicating acquisition. In discrete planning periods, write

$$B_t = B_{\mathrm{loc}}(t) + g_X X_t, \qquad X_{t+1} = X_t + x_t, \qquad g_X \geq 0, \tag{41}$$

where $B_{\mathrm{loc}}(t)$ is non-decreasing with limit $\bar{B} > 0$. Only the replicating channel enters X; observation alone does not. With no previous acquisition, an all-inward history has $X = 0$ and $B_t \leq \bar{B}$. Stopping new investment later does not erase an inherited external stock. Since both stock and local mobilization are non-decreasing, a crossing of B_{tip} cannot be reversed in this model. In a multi-channel model this excludes all-inward expenditure, not necessarily zero replication.

A basin cannot be specified by four scalar parameters without also specifying the resource-mobilization path and the full residual utility. The following benchmark fixes both. It uses one replicating channel, linear residual value, and an already mobilized local ceiling.

Proposition 14 (A unique inward-basin boundary)**.** *Fix $p > 0$, $S > 0$, $r \geq 0$, $d_0 \geq 1$, $g_a > 0$, and $g_X > 0$. Let $X_0 = 0$, $B_{\mathrm{loc}}(t) = \bar{B}$, $a_t = a_0 \exp(g_a t)$, and $0 < \bar{B} < B_{\mathrm{tip}}$, where $(1 + B_{\mathrm{tip}}) \ln(1 + B_{\mathrm{tip}}) = p/S$. At each period choose the unique maximizer*

$$x^*(B, a) = \underset{0 \leq x \leq B/p}{\arg\max} \left\{ \ln[d_0 + a \ln(1 + B - px) + rx] + Sx \right\}. \tag{42}$$

There is a unique $a_{\mathrm{sep}}(g_a, g_X, \bar{B}) \in (0, a^{\ell}_{\mathrm{crit}}(\bar{B}))$ with the following classification:

- *If $a_0 > a_{\mathrm{sep}}$, new external investment stops in finite time and stays zero, at a budget strictly below B_{tip}.*
- *If $a_0 < a_{\mathrm{sep}}$, the budget reaches or exceeds B_{tip} in finite time and new external investment remains positive thereafter.*
- *If $a_0 = a_{\mathrm{sep}}$, every finite-time investment is positive, but $B_t \uparrow B_{\mathrm{tip}}$ and $g_X \sum_{t \geq 0} x_t = B_{\mathrm{tip}} - \bar{B}$.*

Whenever $B_t \geq B_{\mathrm{tip}}$,

$$x_t > \frac{B_t - B_{\mathrm{tip}}}{p}, \qquad B_{t+1} - B_{\mathrm{tip}} > (1 + g_X/p)(B_t - B_{\mathrm{tip}}). \tag{43}$$

The boundary a_{sep} is non-increasing in g_a and non-decreasing in g_X and in $\bar{B}/B_{\mathrm{tip}}$, with the other parameters fixed.

Proof sketch. The unique static investment is continuous, increases with the budget, and decreases with productivity. A path that neither stops nor crosses must converge to the tipping budget. Two such paths cannot coexist: a positive gap between their budgets cannot shrink. Continuity separates the open sets of stopping and crossing paths, giving exactly one boundary. The complete proof and finite-time basin certificates are in section A.7.

Corollary 15 (Saturating productivity and the inward basin)**.** *Retain the one-channel model and parameter assumptions of proposition 14, but replace its productivity path. (i) If a_t is positive, non-decreasing, and bounded by $a_{\max} < a^{\ell}_{\mathrm{crit}}(\bar{B})$, every path reaches B_{tip} in finite time. (ii) Let*

$$a_t = a_{\max} - (a_{\max} - a_0) \exp(-g_a t), \quad 0 < a_0 \leq a_{\max}, \quad a_{\max} > a^{\ell}_{\mathrm{crit}}(\bar{B}). \tag{44}$$

If some initial productivity crosses B_{tip}, a unique $a_{\mathrm{sep}} \in (0, a^{\ell}_{\mathrm{crit}}(\bar{B}))$ separates finite-time capture at higher initial productivity from crossing at lower productivity. Its boundary path has $x_t > 0$ and $B_t \uparrow B^*(a_{\max})$, not B_{tip}. After reaching B_{tip}, equation (43) applies in both cases.

The crossing hypothesis is essential: all paths can instead be captured. The proof and a hard-cap counterexample to the uncaptured-boundary classification are in section A.10.

For example, let $(p, S, r, d_0) = (1, 0.002, 0.05, 1)$, $g_a = 0.05$, $g_X = 0.5$, and $\bar{B} = 0.6B_{\text{tip}}$. New calculations give $B_{\text{tip}} = 106.001289$, $a^{\ell}_{\text{crit}}(\bar{B}) = 7.279612$, and $a_{\text{sep}} \simeq 6.63216$. Starting at $a_0 = 0.5$ crosses the tipping budget at $t = 2$; starting at $a_0 = 50$ is inward from the start. With $a_0 = 0.5$ but $g_X = 0$, investment stops at $t = 54$, not at $t = 0$. Thus removing resource feedback removes this lock-in mechanism, not the initial incentive to expand.

Lock-in here names an absorbing allocation regime, not a welfare mistake. Additional resources do not reduce the feasible set. In this costless, non-depreciating stock benchmark, any finite-horizon continuation value is non-decreasing in X: this follows by backward induction because a larger stock permits the same current project and more local investment. A strictly improving myopic external deviation therefore remains improving after adding such a continuation. A forward-looking planner need not avoid the tipping budget. Productive local alternatives require the separate dynamic comparison developed in the infinite-horizon subsection.

Speculative physical mapping (conjecture). If a fixed-horizon energy budget increases with usable host luminosity, and if denser stellar neighbourhoods reduce external resource prices while leaving residual values fixed, both changes raise $\bar{B}/B_{\text{tip}}$. The basin comparison then suggests a greater propensity to expand. The conjecture concerns these physical parameter mappings, not the proved comparative static. With bounded efficiency, proposition 15 adds the effective ceiling and crossing qualification.

4.5 An infinite-horizon benchmark

Myopic resource acquisition and optimal intertemporal allocation answer different questions. To obtain a transparent infinite-horizon benchmark, replace logarithmic production by linear production, omit the additive computational baseline, and use logarithmic value of strictly positive computation. Let

$$z_t + px_t = B_t, \quad C_t = a_L z_t + a_X x_t, \quad B_{t+1} = g_L z_t + g_X x_t, \quad \max \sum_{t=0}^{\infty} \beta^t \ln C_t. \tag{45}$$

Here $B_0, p, a_L, a_X, g_L, g_X > 0$ and $0 < \beta < 1$. All these parameters are constant. This benchmark is homothetic; it is not the previous logarithmic production model under different notation. For $s = px/B$, define $A(s) = a_L(1 - s) + (a_X/p)s$ and $G(s) = g_L(1 - s) + (g_X/p)s$.

Proposition 16 (Infinite-horizon verification and the inward policy)**.** *Every feasible policy has absolutely convergent discounted log returns. Writing*

$$H_\beta(s) = (1 - \beta) \ln A(s) + \beta \ln G(s), \qquad H^*_\beta = \max_{0 \le s \le 1} H_\beta(s), \tag{46}$$

the value function is

$$V(B) = \frac{\ln B}{1 - \beta} + \frac{H^*_\beta}{(1 - \beta)^2}. \tag{47}$$

A constant maximizing share is optimal forever. The all-inward policy is optimal if and only if

$$(1 - \beta)\frac{a_X/p}{a_L} + \beta \frac{g_X/p}{g_L} \le 1. \tag{48}$$

For every feasible path, $\beta^T V(B_T) \to 0$; this transversality condition validates the Bellman telescoping argument despite unbounded logarithmic returns.

Proof sketch. Positive production and growth endpoints bound $|\ln B_t|$ by a linear function of t. Discounting therefore controls both tails. Substitution of equation (47) into the Bellman equation leaves the one-share maximization in equation (46); its derivative at zero gives equation (48). The complete verification is in section C. The argument supplies its own growth bounds, rather than invoking a bounded-reward contraction. It is consistent with the homogeneous-function approach of Alvarez and Stokey (1998).

Corollary 17 (Static reversals and the patient limit)**.** *Static inwardness is neither necessary nor sufficient for infinite-horizon inwardness. If $(g_X/p)/g_L < 1$, sufficiently patient planners are inward; if this ratio is greater than one, they are not. At equality, the current computational-return ratio still decides for every $\beta < 1$.*

Proof. With $\beta = 0.9$, return ratios $(0.5, 2)$ give a weighted ratio 1.85: static inwardness fails dynamically. Ratios $(2, 0.5)$ give 0.65: static outwardness reverses. Taking $\beta \uparrow 1$ proves the two strict growth cases. With equal growth ratios, subtracting one leaves $(1-\beta)[(a_X/p)/a_L - 1]$, proving the equality case. □

The same derivative decomposition contains the earlier two-period examples, although their nonlinear production is not a literal special case of equation (45). For a concave differentiable first-period objective $U_1(x)$ and increasing concave continuation V_2, the following short transfer result supplies that exact nesting.

Corollary 18 (A future-adjusted inward condition)**.** *If $W(x) = U_1(x) + \beta V_2(B_{20} + g_L B_1 + (g_X - p_1 g_L)x)$ is defined on $[0, B_1/p_1]$, it is concave and its inward endpoint is optimal exactly when $U_1'(0) + \beta V_2'(B_{20} + g_L B_1)(g_X - p_1 g_L) \leq 0$.*

Proof. Composition with the affine continuation budget preserves concavity. The right derivative at the endpoint has the displayed value, and a concave differentiable function has its maximum there exactly when this derivative is non-positive. Substituting the archived U_1, V_2 yields equation (87) in the supplement. □

The archived threshold $g_X = 0.4091$ and allocation $x = 3.7559$ remain unchanged. Their full construction and figure S3 are supplementary evidence for these reversals.

4.6 Residual value and a local ceiling break homotheticity

Adding Sx_t with $S > 0$ makes the reward scale with B_t, whereas the logarithmic reward changes only additively under resource scaling. The closed form equation (47) then ceases to apply. A sufficient condition for finite values is $\beta \max(g_L, g_X/p) < 1$. If $\beta g_X/p \geq 1$, perpetual all-external investment already has infinite positive residual value; it cannot be used as a finite-value optimization counterexample.

Corollary 19 (An unbounded-budget obstruction survives foresight)**.** *In the finite-value residual model, an all-inward infinite-horizon policy can be optimal from a state B only if*

$$\frac{SB}{p} \leq \frac{1 - (1-\beta)(a_X/p)/a_L - \beta(g_X/p)/g_L}{1-\beta}. \tag{49}$$

Thus such a policy cannot persist along $B_t \to \infty$ with constant positive S and fixed finite coefficients.

Proof sketch. Deviate once and then follow the all-inward policy. The continuation log value has derivative $1/[(1-\beta)B]$. Its contribution plus the current residual derivative gives equation (49); see section C. This is a necessary policy test, not a complete solution to the nonhomothetic Bellman equation. At $\beta = 0$ it is the linear-production logarithmic floor without a shifted baseline.

An explicit local ceiling can instead enter through

$$B_{t+1} = \min\{\bar{B}, g_L(B_t - px_t)\} + g_X x_t. \tag{50}$$

Along an all-inward path this keeps future budgets below $\bar{B}$. The transition is no longer homogeneous. For the shifted-log production and linear-residual model of proposition 11, suppose a finite increasing continuation value exists and $g_X \geq pg_L$. Then a small external deviation never reduces next-period resources. Consequently, at $B \geq B_{\text{tip}}$ the strictly positive current external derivative cannot be reversed by continuation value. Below the tipping budget, future resource acquisition can still overturn the static inward optimum. A direct one-step proof is given in section C. This is the valid connection between the local ceiling and the tipping budget; it does not impose the static basin separator on a forward-looking planner.

5 Which residual value survives

The all-inward corner is a stringent diagnostic, but it is not the correct boundary for a claim about the absence of new replicating lineages. Information can be purchased without resident settlement. A dormant, externally stored backup can provide geographical separation without continuing resource acquisition. This section first prices those substitutes and then replaces a free resilience slope by an explicit expected-value calculation. The resulting distinctions identify which part of the earlier floor is a replication incentive rather than merely an external-service incentive.

5.1 No replication with observation and dormant backups

Add an optional dormant-backup capacity $b_{\text{bk}} \geq 0$ at price $p_b > 0$. It has neither computational production nor descendants. The budget and objective become

$$\begin{aligned} z + p_n x + p_h y + p_m w + p_b b_{\text{bk}} &\leq B, \\ U &= F(C) + (1-q)E(x) + \epsilon_R R(x + y + b_{\text{bk}}) + \epsilon_I I(x + y + w). \end{aligned} \tag{51}$$

Throughout this subsection local production is concave; all other functions retain the concavity assumptions of section 2. The no-replication face has $x = y = 0$; observation and backups are optimized, not fixed at zero. Write a maximizing allocation on this face as (z^*, w^*, b^*), its computation as C^*, and its budget multiplier as $\lambda > 0$. Assume derivatives are finite at this allocation. If $z^* > 0$, then $\lambda = F'(C^*)a\ell'(z^*)$. With $z^* = 0$, the multiplier, rather than an unjustified interior formula, is used below.

Proposition 20 (A no-replication criterion)**.** *A no-replication face optimizer is also a global optimizer exactly when it admits a restricted-problem multiplier λ satisfying*

$$\begin{aligned} F'(C^*)r_n + (1-q)E'(0) + \epsilon_R R'(b^*) + \epsilon_I I'(w^*) &\leq p_n \lambda, \\ F'(C^*)r_h + \epsilon_R R'(b^*) + \epsilon_I I'(w^*) &\leq p_h \lambda. \end{aligned} \tag{52}$$

The non-replicating channels satisfy $\epsilon_I I'(w^*) \leq p_m \lambda$ and $\epsilon_R R'(b^*) \leq p_b \lambda$, with equality in each positive channel. The backup inequality is omitted when backups are unavailable. Strict inequalities in equation (52) *exclude replication in every full optimizer.*

Proof sketch. The restricted first-order conditions already control the local, observation, and backup directions. Adding the two displayed inequalities completes the full concave-program optimality conditions. A supporting hyperplane proves global sufficiency. The converse and endpoint cases are proved in section B.2.

When observation is interior and backups are absent, this gives exactly

$$F'(C^*)r_n + (1-q)E'(0) + \epsilon_R R'(0) \le (p_n - p_m)\lambda,$$
$$F'(C^*)r_h + \epsilon_R R'(0) \le (p_h - p_m)\lambda. \tag{53}$$

Information disappears as a separate replication premium, not as an economic influence: it still determines w^*, the remaining local investment, and λ. For an interior backup but a possibly inactive observation channel, the corresponding subtraction is $p_b\lambda$, while $\epsilon_I I'(w^*)$ stays explicit. If both are interior, the comparison uses the net bundle prices $p_j - p_m - p_b$.

Corollary 21 (Substitute bundles and zero-service corners)**.** *At a no-replication face optimum with $w^*, b^* > 0$, $p_h < p_m + p_b$ rules out no replication even if $r_h = 0$. At equality with $r_h = 0$, replacing equal small amounts of observation and backup by industry preserves value and expenditure, and can produce tied replicating optima. If $w^* = 0$, the correct reduction instead uses $c_I = \epsilon_I I'(0)/\lambda \le p_m$; it does not subtract the full observation price. Analogously an inactive backup uses $c_R = \epsilon_R R'(0)/\lambda \le p_b$.*

Proof sketch. Use complementary slackness in equation (52). A cheaper bundle has a strictly positive replication derivative. The exact exchange and zero-channel cases are established in section B.2.

The bundle result is important for interpreting inexpensive autonomous industry. Replacing colonists by observation instruments does not help if a replicating industrial platform supplies both observation and resilience more cheaply than their standalone substitutes. Conversely, expensive industrial platforms can lose even when information and resilience have positive value. The model consequently does not need to erase those motives; it needs a technologically credible way to meet them without a reproducing platform. Independence from a particular service premium is not independence from the prices of its substitutes.

The distinction between exactly zero and a small positive launch rate is consequential only with a propagation mechanism. A deployed viable seed in a deterministic growth law $N(t) = N(0)\exp(gt)$ with $g > 0$ never becomes equivalent to $N(0) = 0$. A stochastic supercritical lineage instead has a positive survival probability, not guaranteed survival. Finite launch sizes, deployment thresholds, and subcritical replication do not justify the stronger claim that every arbitrarily small investment expands without bound. The population calculation in section 7 uses successful, arriving fronts rather than an investment share as its unit.

5.2 A constructive separation from all-inward persistence

The distinction above can change the asymptotic conclusion, not just a finite-budget classification. In the following construction, physical budgets are supplied exogenously to the sequence of static problems. The proposition does not specify how those resources are acquired.

Corollary 22 (Growing budgets without new replication)**.** *Let $\eta = 1$, $B \to \infty$, $a = B^2$, $q = 1$, $C_0 \ge 0$, and $\ell(z) = \ln(1+z)$. Let $\epsilon_I I(w) = J_I[1-\exp(-w/s_I)]$ and $\epsilon_R R(b) = J_R[1-\exp(-b/s_R)]$, with fixed strictly positive $J_I, J_R, s_I, s_R, p_m, p_b$. External computation is concave with fixed finite initial slopes. If $p_n, p_h > p_m + p_b$, then for all sufficiently large B every full optimizer has $x = y = 0$, and a no-replication optimizer has $w, b > 0$ with*

$$z/B \longrightarrow 1, \qquad w + b = O(\ln B).$$

Nevertheless the all-inward corner is eventually impossible because its information and resilience slopes stay positive.

Proof sketch. The restricted face has positive local investment for large B. Its local multiplier tends to zero, making both service channels interior. Their exponential marginal utilities give service quantities logarithmic in the inverse multiplier, hence $B = z + O(\ln z)$. The remaining replication premium divided by that multiplier is $r_j(1+z)/B^2 \to 0$, smaller than either strictly positive net price. A complete proof is in section B.4.

Remark (The growth construction uses $a = B^2$). Keep the other assumptions of proposition 22, but impose $0 < a \le a_{\max} < \infty$. Then $F'(C^*)/\lambda \to \infty$, so every channel with $r_j > 0$ eventually violates equation (57). At $z^* > 0$ this ratio is $(1 + z^*)/a$. With both services interior and $\Delta_j = p_j - p_m - p_b > 0$, the exact test is $a\ell'(z^*) \ge r_j/\Delta_j$; its ceiling counterpart $a_{\max}\ell'(z^*) \ge r_j/\Delta_j$ is necessary, not sufficient. For $r_j = 0$ the channel passes automatically. The proof, including $z^* = 0$, is in section B.5. This is the same declining-marginal-production budget effect isolated by equation (80).

This construction is a counterexample to transferring the all-inward unbounded-budget exclusion directly to no replication. Non-replicating services absorb a persistent preference rather than requiring its weight to vanish. It also explains why a local ceiling below the original B_{tip} cannot be a universal necessary condition once the decision problem changes. The relevant comparison uses the remaining local budget, the optimized shadow value of resources, and the net service-bundle price. A physical application must still provide a feasible resource history; the construction is a separation of allocation claims, not a mechanism for obtaining unbounded local energy.

The price condition is strict for a reason. If an industrial project costs exactly the sum of its substitute services and has no computational return, the exchange in proposition 21 can leave a continuum of tied choices. An analyst selecting the non-replicating member of that set has shown existence, not suppression of every replicating choice. Conversely, if industry costs less than the bundle, improving local computation cannot repair the negative net price while both substitutes remain interior. This distinction matters at the population step: a positive measure of ties resolved outward can create arrival exposure even where an inward optimizer exists. Strict preference, an explicit tie-breaking rule, or a separate containment constraint is needed to exclude that route.

5.3 Value at risk rather than an exogenous resilience slope

For a precise local calculation, specialize to one candidate backup project x, $C(x) = a\ell(B - px) + G(x)$, $G'(0) = r$, and $C(0) = \Omega > 0$. Set $C_0 = 0$ in this calculation. Let $0 \le \vartheta < 1$ be the probability, over the stated horizon, of a catastrophe that separation can mitigate. It is not a rate in inverse years. Let $\varphi(0) = 0$, $0 \le \varphi \le 1$, and $\varphi'(0) > 0$. A seed backup has $\gamma = 0$; a scale-matched mirror has $\gamma = 1$. Use

$$U_R(x) = \left[1 - \vartheta + \vartheta\varphi\left(\frac{x}{C(x)^\gamma}\right)\right] F_\eta(C(x)). \tag{54}$$

This specifies both the value at risk and the probability of retaining it. Its product form need not be concave.

Proposition 23 (The endogenous-risk floor)**.** *Assume a feasible right neighbourhood of $x = 0$ has $C(x) > 0$ and all functions in equation* (54) *are differentiable there. A necessary condition for $x = 0$ to maximize U_R is*

$$p \ge \frac{\Gamma r}{\Omega} + \frac{\vartheta}{1 - \vartheta}\varphi'(0)\Gamma\Omega^{-\gamma}K_\eta(\Omega), \qquad K_\eta(\Omega) = \frac{F_\eta(\Omega)(1 + \Omega)^\eta}{\Omega}. \tag{55}$$

For $0 \le \eta < 1$, $K_\eta(\Omega) \to (1 - \eta)^{-1}$ as $\Omega \to \infty$. For $\eta = 1$ it equals $(1 + \Omega)\ln(1 + \Omega)/\Omega \sim \ln\Omega$.

Proof sketch. At zero the derivative of $x/C(x)^\gamma$ is $\Omega^{-\gamma}$. Differentiate the entire product and divide by the positive surviving marginal-value factor $(1 - \vartheta)F'_\eta(\Omega)$. The exact formula and asymptotic bounds are in section B.3.

The hazard odds, not the hazard probability itself, enter this exact expected-value comparison. Replacing the odds by ϑ is a small-hazard approximation. For $\eta < 1$, a seed therefore retains an asymptotic floor proportional to Γ even as computational output grows. With bounded prices

and unbounded Γ, necessary persistence requires $\vartheta/(1-\vartheta) = O(\Gamma^{-1})$. At a fixed budget, the limiting necessary ceiling, after the remote-computation term vanishes, is

$$\vartheta \leq \frac{p(1-\eta)}{p(1-\eta)+\varphi'(0)\Gamma}. \tag{56}$$

A mirror instead removes the local first-order risk obstruction at bounded positive budgets as $a \to \infty$, because $K_\eta(\Omega)/\Omega \to 0$ for $\eta \leq 1$. This is a statement about the derivative, not global optimality of the nonconcave risk objective. With logarithmic value, a seed is more demanding: its hazard odds must be controlled relative to $1/(\Gamma \ln \Omega)$. Risks unaffected by separation have $\varphi'(0) = 0$ and generate no such marginal premium.

5.4 What remains at perfect experiential substitution

As $q \to 1$, the settlement-specific term $(1-q)E'(0)$ vanishes when $E'(0)$ is fixed and finite. With interior observation and backups, equation (52) then reduces to

$$F'(C^*)r_j \leq (p_j - p_m - p_b)\lambda, \qquad j = n, h. \tag{57}$$

This reduction follows by substitution and continuity of the displayed coefficient; it does not assume that the face optimizer is unchanged when other parameters vary. Strictly positive net prices and sufficiently small remote returns can support no replication. A negative net price precludes it, while a zero net price requires $r_j = 0$ and may admit ties. The residual-free all-inward condition is recovered when the non-replicating channels themselves have zero value and receive zero investment. These are separate limits: substituting experiences, meeting independent information demand, and geographically protecting valued states are not the same technological achievement.

6 Latency and physical scale

6.1 The relevant clock

A remote project competes with what can be experienced before its returns arrive. Let $T_j^{\mathrm{lag}} > 0$ denote its objective delay and let $\omega(u) > 0$ be subjective years per objective year. With a constant subjective discount rate $\delta \geq 0$, the correct discount exponent is

$$\mathcal{D}_j(t) = \delta \int_t^{t+T_j^{\mathrm{lag}}} \omega(u)\,\mathrm{d}u, \qquad r_j(t) = r_{j0}\exp[-\mathcal{D}_j(t)], \quad S_j(t) = S_{j0}\exp[-\mathcal{D}_j(t)]. \tag{58}$$

The simpler product $\delta\omega(t)T_j^{\mathrm{lag}}$ is exact for a clock constant over the delay, and a lower bound for a non-decreasing clock. A clock measured in processor cycles becomes a subjective clock only under an additional account of how those cycles implement valued experience. Neither high throughput nor parallel world creation alone establishes this identification.

Proposition 24 (Latency sufficient for the logarithmic corner)**.** *Take $\eta = 1$, $\ell(z) = \ln(1+z)$, fixed finite non-negative base returns r_{j0}, S_{j0}, $d_0 \geq 1$, and finitely many directions. Suppose $a(t) \geq a_{\min} > 0$ and $p_j(t) \geq p_{\min} > 0$. With $\mathcal{D}_{\min}(t) = \min_j \mathcal{D}_j(t)$, a sufficient condition for eventual strict all-inward allocation is*

$$\mathcal{D}_{\min}(t) - \ln\{[1+B(t)][1+\ln(1+B(t))]\} \longrightarrow \infty. \tag{59}$$

If $B(t) \geq B_{\min} > 0$ eventually, the expression inside the logarithm can instead be $\Gamma(B) = (1+B)\ln(1+B)$. For a non-decreasing clock and $T_j^{\mathrm{lag}} \geq T_{\min}^{\mathrm{lag}} > 0$, either condition follows from its version with $\mathcal{D}_{\min}$ replaced by $\delta T_{\min}^{\mathrm{lag}}\omega(t)$. For $B(t) \leq \bar{B}$, a fixed sufficient threshold is

$$\mathcal{D}_{\min}(t) > \max\left\{0, \max_j \ln \frac{(1+\bar{B})(r_{j0}+d_0 S_{j0})/a_{\min} + \Gamma(\bar{B})S_{j0}}{p_{\min}}\right\}, \tag{60}$$

where a zero numerator is omitted.

Proof sketch. Insert the discounted coefficients into equation (25); the required price is the undiscounted expression multiplied by $\exp(-\mathcal{D}_j)$. The two terms are uniformly bounded by a constant times the bracket in equation (59). Complete bounds and counterexamples to weaker assumptions are in section D.

Latency can therefore remove the constant-residual premise of the growth obstruction. This is a price-adjusted comparison, not an assertion that remote projects cease to produce physical resources. Increasing local speed and reducing travel delay act in opposite directions. Discounting in objective time replaces $\mathcal{D}_j$ by $\delta_{\mathrm{obj}} T_j^{\mathrm{lag}}$; then raising ω has no effect. The choice of clock is a preference assumption with an observable consequence for the model's allocation, not a conclusion forced by relativity or thermodynamics.

Replication itself can invalidate a finite-return specification. Suppose one descendant generation takes the same subjective delay $D > 0$, has expected reproduction factor $\mathcal{R}_X > 0$, and each descendant yields one unit of payoff at birth. The discounted sum from generations $k \geq 1$ is finite exactly when

$$\mathcal{R}_X \exp(-D) < 1, \qquad \sum_{k \geq 1} \mathcal{R}_X^k \exp(-kD) = \frac{\mathcal{R}_X \exp(-D)}{1 - \mathcal{R}_X \exp(-D)}. \tag{61}$$

This follows from the geometric-series criterion, with divergence also at equality. Physical descendants can grow while discounted value stays finite. Conversely, timing the first launch without accounting for its future descendants can substantially understate the effective return. The finite base r_{j0} in proposition 24 must represent the whole specified payoff stream, or a genuinely finite planning horizon.

6.2 Depletion and resilience are different mechanisms

Proposition 25 (No endogenous price drift on a stationary frontier)**.** *Suppose $p_j(t) = P_j(X(t))$ depends only on cumulative past external investment, with $X(t) = X(t_0)$ along an all-inward path starting at t_0. Then $p_j(t) = P_j(X(t_0))$ throughout that path. Frontier depletion alone cannot supply a growing relative price along it.*

Proof. Substitute the constant argument into each P_j. Historical expansion can determine the level at t_0, but cannot create subsequent drift without further investment or another state variable. □

Corollary 26 (A conditional risk cost of discount-induced abstention)**.** *Suppose the only available mitigation of a catastrophe has physical success probability $\varphi(b) > 0$ for $b > 0$, baseline catastrophe probability $\vartheta > 0$, and no effect before its arrival. An allocation with no such backup retains catastrophe probability ϑ after that arrival date, while a completed backup b reduces it to $\vartheta[1 - \varphi(b)]$. A utility discount can change which allocation is preferred but does not reduce either physical probability.*

Proof. These probabilities follow from the stipulated event tree. Multiplying the valuation of a future outcome by a discount factor does not change the probability assigned to that outcome. □

This is the defensible content of a fragility trade-off. There is no general implication from quietness to fragility when dormant backups, local mitigation, negligible hazards, or information about future hazards are available. Only geographically mitigable, otherwise unaddressed risk belongs in this comparison.

Table 1: Assumed marginal advantages and clock speeds: threshold $\delta = \ln \mathcal{R}_{\text{marg}}/(40\omega)$ per subjective year. The forty-year delay and every return/clock entry are scenarios.

$\mathcal{R}_{\text{marg}}$	$\omega = 1$	$\omega = 10^3$	$\omega = 10^6$
10^3	0.17269	1.7269×10^{-4}	1.7269×10^{-7}
10^{10}	0.57565	5.7565×10^{-4}	5.7565×10^{-7}
10^{20}	1.15129	1.1513×10^{-3}	1.1513×10^{-6}
10^{40}	2.30259	2.3026×10^{-3}	2.3026×10^{-6}

Remark (A finite-lived host). If the planning horizon includes certain terminal loss of the usable host, and only geographical separation mitigates it, the stay-put risk model has $\vartheta = 1$. For positive computation and a feasible backup with $\varphi'(0) > 0$, section B.3 excludes the inward corner. Efficiency can postpone resource insufficiency, not remove this stipulated terminal loss; discounting does not lower its probability (proposition 26).

6.3 An explicit dimensional dictionary

The return relevant to delay is a *dimensionless marginal advantage*, not the bare normalized coefficient r_j/p_j:

$$\mathcal{R}_{\text{marg},j} = \frac{r_{j0}}{p_j a \ell'(B)}, \qquad \mathcal{D}_j \gtrsim \ln \mathcal{R}_{\text{marg},j} \tag{62}$$

for a pure-computation comparison with delayed external output. Residual motives require the full preceding inequalities. Resource acquisition by self-replicating industry can make large returns conceivable (Armstrong and Sandberg, 2013); that work does not provide a calibrated value of this paper's normalized r_j/p_j. A ratio 10^{20} below is a scenario assumption, not an empirical estimate attributed to that source.

For an energy-limited illustration, choose $b_{\text{ref}} = 10^{32}\,\text{J}$ and a planning horizon of one Julian year, 31 557 600 s. Using the IAU nominal solar luminosity $L_\odot^{\text{N}} = 3.828 \times 10^{26}\,\text{W}$ (IAU, 2015), a captured fraction f_{cap} gives

$$\bar{B} = 120.8024928\, f_{\text{cap}} \left(\frac{L_*}{L_\odot^{\text{N}}}\right) \left(\frac{\Delta}{1\,\text{yr}}\right) \left(\frac{10^{32}\,\text{J}}{b_{\text{ref}}}\right). \tag{63}$$

This is a per-horizon flow, not the star's lifetime energy. The scalar budget does not replace independent material, memory, or heat constraints. A fixed ceiling describes a specified era and accessible substrate; stellar evolution need not leave it constant indefinitely.

A distance of four light-years traversed at $0.1c$ has a forty-year travel time by definition of a light-year. This is an illustrative one-way journey, not a complete replication cycle; manufacture, deceleration, and return information add delays. For $\mathcal{R}_{\text{marg}} = 10^{20}$, $\omega = 10^6$, and a forty-year delay, equation (62) requires $\delta \gtrsim 1.1513 \times 10^{-6}$ per subjective year. Neither the assumed clock speed nor the discount rate is measured for an extraterrestrial civilization. Their separate roles remain visible in table 1 rather than being buried in a normalized speed parameter.

To interpret table S1 physically, specify b_{ref}, c_{ref}, a project-capacity unit for each direction, and the welfare normalization. Then $B_{\text{res}} = B b_{\text{ref}}$, $C_{\text{useful}} = C c_{\text{ref}}$, a unit project costs $p_j b_{\text{ref}}$, and $r_j c_{\text{ref}}$ is its horizon computation. The local marginal product is $a\ell'(B)c_{\text{ref}}/b_{\text{ref}}$. Benefit amplitudes and scales require an independent valuation model; astronomical dimensions do not identify them. Thus the original normalized interval $B \in [1, 100]$ can represent 10^{32}–$10^{34}\,\text{J}$ under this dictionary, but it is not intrinsically that energy range. Changing the dictionary changes the interpretation, not the frozen experiment.

7 From allocation to silence

7.1 Arrival exposure and the uniformity requirement

An allocation rule is a statement about one decision process. A Fermi explanation requires a distribution of such processes, their launch histories, and propagation to the region where absence is being explained. Let $N_i^{\mathrm{act}}(t)$ denote outward-capable autonomous nodes and $\lambda_i(t)$ their initiation rates per node-time. An independent Poisson approximation gives a surviving-front mean

$$\mu = \sum_i \int \lambda_i(t) N_i^{\mathrm{act}}(t)[1 - q_{R,i}(t)]\,\mathrm{d}t. \tag{64}$$

Here q_R is an extinction probability. For a Galton–Watson process it is the smallest fixed point of the offspring generating function; for Poisson offspring of mean R_{rogue}, $q_R = \exp[R_{\mathrm{rogue}}(q_R - 1)]$ (Harris, 1963). Except for degenerate immortal one-descendant processes, a mean at most one gives extinction almost surely. Survival is not arrival, and extinct fronts can still leave finite artefacts. Replace μ by $\mu_A(z, T)$ when launch, survival, and propagation into the target region by time T have all been accounted for.

For a fixed independent list of civilizations with Poisson arrival means $\mu_{A,i}$, $P_0^{\mathrm{fixed}} = \exp(-\sum_i \mu_{A,i})$. If origins themselves form a Poisson process with intensity measure $\nu(\,\mathrm{d}z)$, independently marked by their arrival means, the void probability instead is

$$P_0^{\mathrm{origins}} = \exp\left[-\int (1 - \exp[-\mu_A(z, T)])\nu(\,\mathrm{d}z)\right]. \tag{65}$$

A random number of parent civilizations clusters fronts and changes the likelihood. For two fixed civilizations with mean two fronts each the probability is 0.0183; for a Poisson number of mean two it is 0.1774. These archived numbers are retained. The derivation and a new simulation are given in section E.

Proposition 27 (Necessary uniformity and the exact two-type window)**.** *Let* $0 < \bar{N}_A = \int \nu(\,\mathrm{d}z) < \infty$, $0 < P_* < 1$, *and* $\mathcal{L}_* = -\ln P_*$. *Suppose a fraction* π_{out} *of origins under the normalized arrival-relevant intensity has* $\mu_A \geq \mu_{\mathrm{out}} > 0$. *Then* $P_0 \geq P_*$ *requires*

$$\pi_{\mathrm{out}} \leq \frac{\mathcal{L}_*}{\bar{N}_A[1 - \exp(-\mu_{\mathrm{out}})]}. \tag{66}$$

For the complementary class labelled inward, let $c_{\mathrm{in}} = \mathbb{E}[1 - \exp(-\mu_A) \mid \textit{inward}]$. *Set* $c_{\mathrm{in}} = 0$ *when that class has zero probability. Then also* $(1 - \pi_{\mathrm{out}})c_{\mathrm{in}} \leq \mathcal{L}_*/\bar{N}_A$. *For two types with exactly specified means* $\mu_{\mathrm{out}} > \mu_{\mathrm{in}} \geq 0$, *write* $f_k = 1 - \exp(-\mu_k)$. *The exact necessary and sufficient condition is*

$$f_{\mathrm{in}} + \pi_{\mathrm{out}}(f_{\mathrm{out}} - f_{\mathrm{in}}) \leq \mathcal{L}_*/\bar{N}_A. \tag{67}$$

Without any upper bound on the means, $\bar{N}_A \leq \mathcal{L}_*$ *is sufficient for* $P_0 \geq P_*$. *The weaker condition* $\bar{N}_A f_{\mathrm{out}} \leq \mathcal{L}_*$ *is sufficient for arbitrary mixtures only in the exactly specified two-type model, not when* μ_{out} *is merely a lower bound.*

Proof sketch. The negative logarithm of equation (65) is $\bar{N}_A \mathbb{E}[1 - \exp(-\mu_A)]$. Bound its non-negative outward and inward contributions separately; evaluate the expectation exactly for two types. A complete proof and boundary cases are in section E.

At $P_* = 0.05$ and $\mu_{\mathrm{out}} = 3$, the necessary upper outward fractions for $\bar{N}_A = 10^2, 10^3, 10^4$ are respectively 3.1527%, 0.31527%, and 0.031527%. These are conditional compatibility bounds, not posterior probabilities. For example, a prior on the number of origins could make large exposure unlikely; then uniform inward behaviour need not be invoked. Conversely, with a large arrival-relevant population, a small expansionist tail can exhaust the entire permitted exposure. Any inward leakage makes the constraint on that tail tighter, not looser.

The archived design is far from a near-uniform all-inward allocation: its reference count is 219/1024, and its design-measure sensitivity is reported in the supplement. But 805 failed all-inward tests cannot be identified with 805 replicating civilizations. Reoptimizing observation on the no-replication face, with exactly the same frozen inputs and no new dormant-backup technology, gives 421/1024 no-replication optima. Of these, 202 use observation without replication; 603/1024 prefer replication. These additive calculations strengthen the logical distinction rather than providing an astronomical prevalence estimate. Even this selected synthetic ensemble is nowhere near the high-exposure uniformity requirement under a literal population interpretation, which its sampling design does not justify.

Under a uniform external price multiplier κ, the archived statistic ρ gives $\Pr(\rho > \kappa)$ for loss of the *all-inward* corner. It equals an outward population fraction only after specifying a population measure and identifying every such failure with the relevant outward behaviour. With observation or backups free, the face optimizer must be recalculated; the old ρ is not the no-replication threshold. This is why a change in wording alone cannot carry the old sensitivity experiment into the new population claim.

7.2 Preemption: a game rather than an adjusted corner

To formalize strategic complementarity, let $N \geq 2$ civilizations choose $x_i \in [0, \bar{X}]$. Suppose

$$U_i = f(x_i) + \iota x_i \sum_{k \neq i} x_k, \qquad \iota \geq 0, \tag{68}$$

where f is continuously differentiable and strictly concave. It can include the static local-versus-external objective, including the base residual slope S_{j0}. This is a specified payoff externality, not a claim that fear of neighbours has a universal linear form. Competition among expanding lineages is an established motivation for such a test (Hanson, 1998).

Proposition 28 (Preemption equilibria and a separate cascade test)**.** *In equation* (68)*, a profile is a pure Nash equilibrium exactly when, with* $X_{-i} = \sum_{k \neq i} x_k$*,*

$$\begin{cases} f'(0) + \iota X_{-i} \leq 0, & x_i = 0, \\ f'(x_i) + \iota X_{-i} = 0, & 0 < x_i < \bar{X}, \\ f'(\bar{X}) + \iota X_{-i} \geq 0, & x_i = \bar{X}. \end{cases} \tag{69}$$

A pure equilibrium exists. The all-inward profile is an equilibrium iff $f'(0) \leq 0$*; the all-cap profile is an equilibrium iff* $f'(\bar{X}) + \iota(N-1)\bar{X} \geq 0$*. When both hold there are at least two equilibria. Symmetric interior equilibria are exactly the roots of* $f'(x) + \iota(N-1)x = 0$ *in* $(0, \bar{X})$*.*

For the distinct binary-action game $x_i \in \{0, \bar{X}\}$*, put* $d_{\text{bin}} = [f(\bar{X}) - f(0)]/\bar{X}$*. Its all-inward and all-outward profiles are equilibria iff, respectively,* $d_{\text{bin}} \leq 0$ *and* $d_{\text{bin}} + \iota(N-1)\bar{X} \geq 0$*. With inward tie-breaking, one externally maintained outward player triggers all other players under asynchronous best-response updates iff* $d_{\text{bin}} + \iota\bar{X} > 0$*.*

Proof sketch. Strict concavity gives the endpoint and interior derivative conditions. The potential $\sum_i f(x_i) + \iota \sum_{i<k} x_i x_k$ has a maximum on the compact strategy cube. Binary choices instead compare finite payoff differences; they cannot be tested using $f'(0)$ alone. Full proofs are in section E.1.

Failure of the inward derivative test at maximal rival investment does not establish an outward equilibrium. For $N = 2$, $\bar{X} = 1$, $f(x) = -x - 5x^2$, and $\iota = 2$, that derivative is $-1 + 2 = 1 > 0$, but the only equilibrium is $(0, 0)$. Nor do two stable endpoint equilibria imply a one-defector cascade. In the binary game with five players, $d_{\text{bin}} = -2.5$ and $\iota = \bar{X} = 1$, both endpoints are stable, but a single defector gives the remaining players incentive $-1.5 < 0$. A sufficiently large cluster can matter when one player does not. Population uniformity is

therefore an equilibrium-selection and containment problem as well as a distribution of individual preferences.

Virtual population growth affects that problem only through physical agency. If worlds grow as $\exp(g_W t)$, their outward-capable fraction falls as $\exp(-g_c t)$, and the launch-times-arrival factor stays bounded above and below by positive constants, integrated infinite-horizon exposure is finite exactly when $g_c > g_W$. This is the elementary integrability criterion for $\int_0^\infty \exp[(g_W - g_c)t]\,\mathrm{d}t$. Without actuator access the exposure can instead be zero, regardless of the virtual population. Recursive population count alone is neither a launch rate nor a guarantee of containment.

7.3 Visibility is a separate inverse problem

Neither an allocation share nor an arrival probability fixes spectrum, power, geometry, or duty cycle. With mutually exclusive source classes, a schematic Poisson detection model is

$$\Lambda_j(T) = \int_0^T R_{\mathrm{civ}}(t_e)\mathcal{P}_j(T - t_e \mid t_e)\pi_j(T, t_e)\,\mathrm{d}t_e, \qquad P(0) = \exp[-\sum_j \Lambda_j(T)]. \tag{70}$$

Here $\mathcal{P}_j$ is a source-state probability, not the residual slope S_j, and π_j is the selection probability. Overlapping source attributes require union probabilities rather than double counting. Even a stationary one-class model constrains a product of emergence rate, source lifetime τ, and selection probability. Increasing one factor and decreasing another can preserve the likelihood. A null survey therefore does not by itself identify an inward transition.

7.4 A limited thermal bridge

Landauer's bound assigns at least $k_B T \ln 2$ of heat to erasure of one unbiased bit under its thermodynamic assumptions (Landauer, 1961). It is not a universal energy charge per reversible logical operation or a law fixing subjective clock speed (Bennett, 1973). For an ideal one-temperature blackbody radiator, negligible background temperature, total effective emitting area $A = M_{\mathrm{rad}}/\Sigma_{\mathrm{rad}}$, and thermalized luminosity L, the Stefan–Boltzmann law gives

$$T = \left(\frac{L\Sigma_{\mathrm{rad}}}{\sigma_{\mathrm{SB}} M_{\mathrm{rad}}}\right)^{1/4}. \tag{71}$$

This equation follows by solving $L = \sigma_{\mathrm{SB}} A T^4$ for T; A already counts all radiating faces. With nominal solar luminosity, $\Sigma_{\mathrm{rad}} = 1\,\mathrm{kg\,m^{-2}}$ as an assumed engineering parameter, and the JPL planetary masses (JPL, 2026), a Mercury mass gives 378.16 K and a Jupiter mass 43.43 K. Their wavelength-form Wien peaks are 7.66 and 66.73 μm, respectively. Constants and numerical root checks are specified in the supplement (NIST, 2024).

The first example lies inside the 100–700 K model range searched by Project Hephaistos II; the second lies below it (Suazo et al., 2024). The WISE/Spitzer framework of Wright et al. (2014) likewise targets mid-infrared waste heat, not all possible cold radiator architectures. Thus inward allocation does not predict a unique cold far-infrared population. It permits temperature-dependent search targets once dissipated power and radiating area are supplied. Bradbury's Matrioshka-brain manuscript already describes nested stellar computing and heat disposal (Bradbury, 1999). The cooler galactic environments of Ćirković and Bradbury (2006) can motivate migration, not necessarily continued residence around the original star. The present bridge supplies no priority claim for either idea and no construction-feasibility proof.

8 Verification and limitations

8.1 What the calculations verify

The mathematical results are proved under the assumptions printed with their statements. Numerical checks then test equations, implementations, examples, and reductions; they do

not supply evidence that extraterrestrial parameters satisfy those assumptions. The archive is retained unchanged. Re-execution of its scripts reproduces the original classifications, threshold tests, workload examples, and two-period checks. New scripts independently evaluate full objectives or explicit dynamics rather than merely calling the formulas being tested. Scalar global searches are used for the nonconcave exponential examples; concave allocation checks use analytic derivatives and constrained optimization. A finite-horizon Bellman recursion is compared with the infinite-horizon value formula, and direct quadrature checks the changing-clock integral.

The new no-replication design result is additive. It does not replace the original all-inward counts or silently change their meaning. Separate outputs record the restricted face optimum, the completed replication inequalities, and sampled full-objective comparisons. The selected thermal constants have a dimensional provenance, whereas the assumed clock speeds, marginal advantages, and radiator areal density are labelled as scenarios. Code fixes during verification are followed by complete reruns of the affected script. Recorded pass counts refer to assertions, not necessarily independent samples or experimentally observed systems.

8.2 Consolidated limitations

Physical realizability and the workload margin. No theorem proves that arbitrarily large subjective populations fit inside a real stellar system. Efficiency, cooling, fidelity, error correction, material replacement, and stored-state capacity can plateau. Reversible logic does not remove the costs of preparation, control, or access to entropy reservoirs. The world-count lemma derives curvature only after choosing an aggregation ethic, feasible per-world resources, and a margin for varying the number of worlds. Recursion is one implementation of that margin, not its exclusive cause. An embodied civilization using virtual experiences requires less than a civilization implementing conscious digital agents; the model does not establish either engineering capability or a theory of consciousness.

The boundary is model-specific. Concavity, divisibility, finite derivatives at tested corners, and positive prices make local derivatives globally informative. Fixed project sizes, start-up costs, and some increasing returns do not. The exponential endpoint lemma states an exception and its counterexamples. Likewise, endogenizing resilience produces a product objective for which the derived floor is only necessary. The fixed-residual tipping budget is not invariant when observation and backups are reoptimized, a hazard changes with scale, or prices change with geography. Its role is to expose an obstruction in the stated benchmark, not to assign every civilization a universal budget number. Equality can admit multiple optima; no-replication as a selected optimum is weaker than exclusion of all replicating optima.

Dynamics and inherited infrastructure. The basin theorem uses a non-depreciating stock, a specified productivity path, linear residual utility, and myopic choices. Its exact separatrix is not an optimal infinite-horizon stopping rule. The homothetic benchmark has different technology and proves its own infinite-horizon policy; the ceiling comparison gives only a one-sided connection to the static tipping budget. A linear residual payoff can make discounted value infinite unless growth and discounting are controlled. More general persistent capital, uncertainty, strategic resource exclusion, and option values need additional state variables. Stopping new investment leaves previous autonomous projects in place. Their future descendants must be included in arrival exposure rather than erased by relabelling the parent civilization inward.

Timing, geography, and welfare. Subjective-time discounting is optional, not a physical requirement. Parallel worlds need not run faster, and even a fast internal clock need not imply impatience about external opportunities. Delays can devalue resilience as well as replication. Only hazards reduced by separation contribute to the geographical backup premium; a seed

and a mirror also have different restoration capabilities beyond the model's scale exponent. If energy efficiency is obtained through slower adiabatic or reversible computation, a and ω are coupled: raising a can lower ω and weaken the delay mechanism in proposition 24. Quantifying this trade-off is left to future work.

The luminosity and neighbourhood-density interpretation of the basin is explicitly conjectural. The normalized access map in the supplement is not a propulsion law, and its superluminal limits are sensitivity counterfactuals, not physically established travel mechanisms.

Population inference and observation. The synthetic ensemble is not a prior justified by astronomical data. An origin fraction is not a fraction of occupied systems. The Poisson model assumes independent marking and a specified arrival exposure; correlated origins, interacting fronts, and selection effects can change its likelihood. The strategic game establishes possibilities of coordination and cascades under particular actions and update rules, not a measured prevalence of preemption. Survey non-detections constrain emergence, lifetime, and selection jointly. A radiator temperature does not fix a detection probability without distance, spectrum, dust, covering fraction, duty cycle, and survey completeness. Neither a warm candidate nor a cold non-detection selectively confirms CITH.

9 Conclusion

The default result of the logarithmic benchmark is outward pressure from growing resources, not automatic retreat into virtual worlds. Concave local production has a residual-value floor at least proportional to the physical budget. With persistent external motives and bounded prices, no increase in computational productivity alone defeats that obstruction. A bounded local resource base changes the question: below a unique tipping budget, sufficient productivity can support inward allocation, subject to the lower $B^*(a_{\max})$ when efficiency is capped; at or above B_{tip}, no finite productivity can. In the exponential-productivity acquisition benchmark, premature expansion can cross that boundary before local efficiency catches up, and a unique critical initial productivity separates the paths. A saturating path requires the crossing qualification in proposition 15.

A Fermi-relevant explanation must then replace the all-inward corner by the absence of new replication. Observation and dormant backups can absorb external-service demand, but a cheaper reproducing bundle can defeat them. When resilience protects the value accumulated internally, a seed backup retains a strong marginal motive rather than becoming irrelevant merely because computers improve. Value curvature and the treatment of world number determine whether productivity is capable of overcoming these pressures. Future resource production and the timing of remote returns supply further tests; neither is represented by a static capability to simulate appealing experiences.

Thus a bounded-resource, substitute-service, latency-based inward scenario must clear three distinct comparisons. Other preference and technology regimes can take different routes. Even a successful individual scenario becomes a silence explanation only after meeting the population constraint: when the arrival-relevant origin population is large, the permitted expansionist tail and leaked physical agency must be extremely small. When that population is small, rarity may suffice without an inward transition. The theory identifies the parameters and inequalities on which these alternatives disagree. It supplies neither an estimated civilizational prevalence nor a general expectation of cold, invisible computers. Its explanatory value is the ability to state where an inward mechanism survives, where it fails, and what additional evidence would be needed to distinguish it from expansion or simple rarity.

Declarations

Data and code availability. The accompanying package contains this complete source, the manuscript and supplement, the unchanged v7.3 synthetic inputs and figures, all new validation scripts and outputs, and an integrity manifest. The README gives reproduction commands. No newly assigned repository DOI is asserted in this version.

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Use of artificial-intelligence tools. The critique, proposed extensions, and initial numerical spot checks were produced using Anthropic's Claude. OpenAI ChatGPT drafted and integrated the v8 material, developed proofs and counterexamples, wrote and executed additional numerical checks, and assisted bibliographic and LaTeX verification. Both the generating and checking implementations are AI-assisted; separate implementations are not independent human peer review. No AI system is an author. The human author retains responsibility for approval of all claims and the submission decision.

Ethics. No human participants, animals, personal research data, or identifiable-person observational data were studied.

A Proofs for the corner and persistence results

A.1 Tightness of the production floor

Proof of corollary 5.1. The supporting-line inequality for a concave function, based at B, reads $\ell(v) \le \ell(B) + \ell'(B)(v - B)$ for $0 \le v \le B$. At $v = 0$ it gives $B\ell'(B) \le \ell(B)$, proving $\Gamma(B) \ge B$. If equality holds, concavity also gives $\ell(v) \ge (v/B)\ell(B) = v\ell'(B)$. The supporting line is $v\ell'(B)$, so both inequalities are equalities throughout $[0, B]$. Conversely a linear function on that interval plainly gives equality. No assertion about production beyond B is needed.

At $\eta = 1$, equation (18) becomes

$$p_j \ge S_j\Gamma(B) + \frac{r_j + d_0 S_j}{a\ell'(B)}.$$

The second term is strictly positive whenever $S_j > 0$, since $d_0 > 0$ and $a\ell'(B) > 0$. Thus $p_j > S_j B$ is necessary in every such direction for any admissible concave production function. If those inequalities hold, choose $\ell(v) = v$. All denominators $p_j - S_j B$ are positive, including p_j in a zero-slope direction. The finite choice

$$a > \max_j \frac{r_j + d_0 S_j}{p_j - S_j B}$$

satisfies every corner inequality strictly. A direction with $r_j = S_j = 0$ contributes zero to this maximum. If all directions are trivial, any positive a works. This proves necessity and sufficiency of the claimed existence condition, including zero residual slopes. Along an unbounded-budget path, a fixed positive slope thus requires $p_j(t) > S_j B(t)$. If $p_j(t) \le M$, then $S_j(t) < M/B(t)$ at positive-slope times, and hence $S_j(t) = O(B(t)^{-1})$ including zero-slope times. These are necessary rate restrictions; the remaining productivity term is still required. □

A.2 Growing budgets and the original special cases

Proof of proposition 8. Multiplying equation (18) by Γ gives the required price

$$P_{j,\eta}(\Omega, B) = \Gamma(B)\left[\frac{r_j}{\Omega} + S_j \frac{(d_0+\Omega)^\eta}{\Omega}\right].$$

For $\eta = 1$, the second quotient is $1 + d_0/\Omega > 1$. Hence a passing direction with $p_j \le M$ and $S_j \ge s$ requires $M > s\Gamma(B) \ge sB$, which eventually fails. For $\eta > 1$, differentiation gives

$$\frac{\mathrm{d}}{\mathrm{d}\Omega}\frac{(d_0+\Omega)^\eta}{\Omega} = \frac{(d_0+\Omega)^{\eta-1}}{\Omega^2}[(\eta-1)\Omega - d_0].$$

The quotient diverges at both endpoints of $(0,\infty)$, decreases until $\Omega = d_0/(\eta-1)$, and increases thereafter. Substitution gives equation (33). Consequently the required price is at least sB times a fixed positive number and again eventually exceeds M. The argument applies separately at each time and imposes no regularity or growth restriction on $a(t)$.

For $0 \le \eta < 1$, $(d_0+\Omega)^\eta \ge \Omega^\eta$. A passing direction therefore requires $M \ge s\Gamma\Omega^{\eta-1}$, which is exactly equation (30). For sufficiency, fix the parameters as in the proposition, let $0 < \eta < 1$, and suppose $\Omega \ge c_\Omega \Gamma^{1/(1-\eta)}$. Since $\Gamma \ge B \to \infty$, also $\Omega \to \infty$, and

$$\begin{aligned} P_{j,\eta} &= \frac{r_j\Gamma}{\Omega} + S_j\Gamma\Omega^{\eta-1}(1+d_0/\Omega)^\eta \\ &\le \frac{r_j}{c_\Omega}\Gamma^{-\eta/(1-\eta)} + S_j c_\Omega^{\eta-1}(1+d_0/\Omega)^\eta. \end{aligned}$$

The first term tends to zero. The limsup of the second is strictly less than p_j by the stated strict bound on c_Ω; when $S_j = 0$ both terms tend to zero. Finitely many directions admit a common time beyond which all inequalities are strict. For $\eta = 0$ the same exact price is $(r_j + S_j)\Gamma/\Omega$, proving equation (32). Finally, $\Gamma = B$ and $\Omega = aB$ for linear production give the asserted rate for a. □

The sufficiency assumptions cannot be replaced by an upper price bound and a lower residual bound alone. For example, take $\eta = 1/2$, $\ell(z) = z$, $d_0 = 1$, $r = 0$, $S = 1$, $B(t) = t$, $a(t) = 10t$, and $p(t) = \exp(-t)$. This productivity has the proposed power-law rate, but its required price tends to $1/\sqrt{10}$ while its actual price tends to zero. Likewise equality in the asymptotic coefficient is not generally enough: with $\ell(z) = z$, $S = p = d_0 = 1$, $r = 0$, and $a = B^{\eta/(1-\eta)}$, the required price is $(1+1/\Omega)^\eta > 1$ for every finite B and $0 < \eta < 1$.

For clarity, the older reductions are algebraic, not additional assumptions. Substituting $\eta = 1$ gives equation (22); additionally setting $\ell(B) = \ln(1+B)$ gives H_j, A_j and equation (25). Keeping arbitrary η with this production function gives equation (40). Applying the logarithmic formula at each time gives equation (34); its zero-A_j case is exactly $r_j = S_j = 0$. The general production floor grows at least linearly, whereas the logarithmic special case is $(1+B)\ln(1+B) \sim B\ln B$.

The original pure-computation solution also follows. If all residual weights vanish and external computation is affine, any strictly increasing F has the same optimizer as C. At a fixed local allocation z, remaining resources are assigned to a channel with $R_* = \max(r_n/p_n, r_h/p_h)$. For $R_* > 0$, the reduced derivative is $a/(1+z) - R_*$ and is strictly decreasing. Clipping its root to $[0, B]$ gives $z^* = \min\{B, \max(a/R_* - 1, 0)\}$, which is equation (80). If $R_* = 0$, positive local marginal production uses the whole budget. Ties between best external channels affect the composition of the remainder, not the local allocation.

A.3 Increasing returns and an exceptional global test

Proof of proposition 10. The necessary logarithmic test, which does not require concavity of ℓ, implies $p_j > S_j \ell(B)/\ell'(B)$ wherever $S_j > 0$. Therefore $\ell'(B(t))/\ell(B(t)) > S_j(t)/p_j(t) \geq s/M$ at every sufficiently late time. Choose such a time t_0. Continuity and $B(t) \to \infty$ imply that every $B > B(t_0)$ is attained at some $t \geq t_0$, by the intermediate value theorem. Thus the strict differential inequality holds throughout that budget interval. Integrating it on any $[B_0, B]$ contained there gives $\ln \ell(B) - \ln \ell(B_0) > (s/M)(B - B_0)$. For convex ℓ with $\ell(0) = 0$, the tangent inequality at B is $0 \geq \ell(B) - B\ell'(B)$, so $\Gamma \leq B$. Direct differentiation gives $\Gamma = B/k$ for $\ell(z) = z^k$ and $\Gamma = [1 - \exp(-\psi B)]/\psi$ for the exponential specification. Every fixed finite k still yields $\Gamma \to \infty$, contradicting the strict price floor when prices are bounded and residual slopes stay positive. □

Lemma 29 (Exponential production: an exact endpoint test)**.** *Let $B, a, p, \psi > 0$, $d_0 \geq 1$, and $r, S \geq 0$. On $[0, B/p]$, define*

$$U(x) = \ln\left[d_0 + \frac{a}{\psi}\{\exp[\psi(B - px)] - 1\} + rx\right] + Sx. \tag{72}$$

Then $x = 0$ is a global maximizer if and only if $U'(0) \leq 0$ and $U(0) \geq U(B/p)$. The result allows ties and does not extend to arbitrary concave residual utility or arbitrary concave external computation.

Proof. Necessity follows from a feasible right derivative and endpoint comparison. For sufficiency, put $k = \psi p$, $A = (a/\psi)\exp(\psi B)$, and $c = d_0 - a/\psi$. The positive argument of the logarithm is $D(x) = A\exp(-kx) + rx + c$. The derivative has the sign of

$$f(x) = D(x)U'(x) = (S - k)A\exp(-kx) + Srx + r + Sc.$$

If $S \leq k$, then $f'(x) = k(k - S)A\exp(-kx) + Sr \geq 0$. Hence $f(0) \leq 0$ makes the non-positive set an initial interval. If $S > k$, then $f''(x) = k^2(S - k)A\exp(-kx) > 0$. Its non-positive sublevel set is convex and contains zero, so is again an initial interval, possibly a singleton or the entire domain. In both cases U is non-increasing and then non-decreasing. Its maximum is consequently attained at an endpoint. The assumed endpoint comparison proves the result. For $r = 0$, $a = \psi d_0$, and $S = \psi p$, the function is constant; thus a strict uniqueness conclusion would be false. □

A.4 Smooth counterexamples to the endpoint shortcut

These constructions use positive logarithm arguments throughout and infinitely differentiable, non-decreasing, concave alternatives. They refute extensions of proposition 29, not its stated affine and linear assumptions. An interior comparison suffices; locating the precise global optimizer is unnecessary.

First set $B = 10$, $p = \psi = d_0 = 1$, $a = 0.01$, and $r = 0$, and replace Sx by

$$V_{\text{ext}}(x) = \frac{0.76}{20}\left\{\ln(1 + \exp(140)) - \ln(1 + \exp[20(7 - x)])\right\}.$$

Here $V_{\text{ext}}(0) = 0$, $V'_{\text{ext}}(x) = 0.76/[1 + \exp(20(x - 7))] > 0$, and $V''_{\text{ext}}(x) < 0$. Let $U(x) = \ln[1 + 0.01\{\exp(10 - x) - 1\}] + V_{\text{ext}}(x)$. The following strict intervals are sufficient:

$$-0.236 < U'(0) < -0.235, \qquad 5.399 < U(0) < 5.400,$$
$$5.468 < U(7) < 5.469, \qquad 5.319 < U(10) < 5.321.$$

Thus the first-order condition and comparison with the all-external endpoint both pass, but the interior value at $x = 7$ is larger. These bounds can be checked directly from the displayed

expressions, using $\ln(1+\exp v) = v + \ln(1+\exp(-v))$ for $v \geq 0$ and $0 < \ln(1+u) < u$ for $u > 0$; the gaps exceed 0.06. A looser analytic certificate already suffices: $U(0) < 5.4$, $U(7) > 0.17 + 0.038(140 - 0.694) > 5.463$, and $U(10) < 5.32 < U(0)$. These follow from $\exp(3) > 20$, $\ln(1.19) > 0.17$, $\ln 2 < 0.694$, $\exp(10) < 22027$, $\exp(5.4) > 221.4$, and $\ln 100 < 4.606$. Furthermore, the negative computational derivative has magnitude greater than 0.99 while the residual derivative is less than 0.76, so $U'(0) < 0$. The numerical optimizer is an additional check, not the argument for the counterexample.

Second set $B = 3$, $p = \psi = a = d_0 = 1$, and $S = 0$, and replace affine external production by

$$G(x) = \frac{9}{50}\left\{\ln(1+\exp(100)) - \ln(1+\exp[50(2-x)])\right\}.$$

Then $G(0) = 0$, $G'(x) = 9/[1+\exp(50(x-2))] > 0$, and $G''(x) < 0$. For $U(x) = \ln[\exp(3-x) + G(x)]$, one has

$$U'(0) = -1 + \frac{9}{\exp(3)[1+\exp(-100)]} < 0, \qquad U(0) = 3.$$

Also $G(3) < 18$, so $U(3) < \ln 19 < 3$, whereas $G(2) > 18 - (9/50)\ln 2$ gives

$$\exp(1) + G(2) > \exp(1) + 18 - (9/50)\ln 2 > \exp(3),$$

so $U(2) > 3$. For example, the elementary bounds $\exp(1) > 2.718$, $\exp(3) < 20.086$, and $\ln 2 < 0.694$ already prove the last strict comparison. The two-test shortcut fails even with zero residual utility when the external production function is allowed to be concave rather than affine.

A.5 Existence and uniqueness of the tipping budget

Proof of proposition 11. The assumed positive derivative makes Γ well defined and continuously differentiable on $[0,\infty)$. The quotient rule gives equation (38). Hence Γ starts at zero, is strictly increasing, and is unbounded because $\Gamma(B) \geq B$. If a positive slope exists, the finite number $\min_{S_j>0} p_j/S_j$ is strictly positive. The intermediate value theorem and strict monotonicity give exactly one positive crossing.

Write $d_j(B) = p_j\ell'(B) - S_j\ell(B) = \ell'(B)[p_j - S_j\Gamma(B)]$. Below the crossing each d_j is positive; above or at it, a direction attaining the minimum has $d_j \leq 0$ and $r_j + d_0 S_j > 0$. The exact logarithmic threshold then proves attainability below and non-attainability at or above the crossing. On the lower interval, $d_j'(B) = p_j\ell''(B) - S_j\ell'(B) \leq 0$. Thus every nontrivial ratio $(r_j + d_0 S_j)/d_j(B)$, and their finite maximum, is non-decreasing. At least one positive numerator has a denominator tending to zero as $B \uparrow B_{\text{tip}}$, proving divergence.

If $B(t) \leq B' < B_{\text{tip}}$ eventually, the threshold is bounded above by $a^{\ell}_{\text{crit}}(B') < \infty$. The revised hypothesis states that $a(t)$ strictly exceeds this finite bound eventually; divergence of productivity is unnecessary. Every nontrivial corner inequality is then strict; in a trivial direction it is also strict because $p_j > 0$ and the local marginal value is positive. The hub proposition therefore makes every optimizer all-inward. If all $S_j = 0$, all thresholds are $r_j/[p_j\ell'(B)]$, finite at every finite budget. There is then no residual tipping budget; the definition $B_{\text{tip}} = \infty$ records this case. □

A.6 Value curvature at fixed and bounded budgets

Proof of proposition 13. Fix $B > 0$ and abbreviate $L_\ell = \ell(B) > 0$ and $d_\ell = \ell'(B) > 0$. A direction passes exactly when

$$P_{j,\eta}(a) := \frac{r_j + S_j(d_0 + aL_\ell)^\eta}{a d_\ell} \leq p_j.$$

For a nontrivial direction and $0 \le \eta < 1$,

$$P'_{j,\eta}(a) = \frac{-r_j + S_j(d_0 + aL_\ell)^{\eta-1}[(\eta-1)aL_\ell - d_0]}{a^2 d_\ell} < 0.$$

The expression tends to infinity as $a \downarrow 0$ and to zero as $a \to \infty$. Continuity and strict monotonicity prove the unique finite threshold. At $\eta = 1$ expansion gives equation (22) and hence equation (23). At $\eta > 1$ and $S_j > 0$, the sign condition is the non-positive sublevel set of

$$r_j + S_j(d_0 + aL_\ell)^\eta - p_j a d_\ell.$$

This function is strictly convex, positive at zero, and tends to infinity as $a \to \infty$. Its non-positive set is therefore empty, a singleton, or a bounded closed interval separated from zero. For $S_j = 0$ the exact inequality is simply $a \ge r_j/(p_j d_\ell)$, with a trivial direction unrestricted. Putting $L_\ell = \ln(1+B)$ and $d_\ell = (1+B)^{-1}$ recovers equation (40) and every fixed-budget statement of the original curvature proposition.

For the bounded-budget conclusions, take a compact positive-budget interval containing all sufficiently late $B(t)$ and the limit $\bar{B} > 0$. On this interval L_ℓ is bounded above and below by positive constants and d_ℓ is likewise bounded away from zero and above. These statements follow from the maintained differentiability and concavity assumptions locally, or directly from the C^2 assumptions of the tipping result. For $\eta < 1$, the preceding required prices are uniformly $O(a^{-1} + a^{\eta-1})$ and hence tend to zero as a function of a. Choose a finite threshold above which these bounds are strictly below every positive p_j; it suffices for the actual path to eventually exceed that threshold. For $\eta > 1$ and $S_j > 0$, they are uniformly bounded below by a positive constant times $a^{\eta-1}$. A finite threshold above which this lower bound exceeds p_j likewise suffices for exclusion. At $\eta = 1$, a limiting budget below the tipping point is eventually bounded by some $B' < B_{\text{tip}}$, and eventual $a(t) > a^\ell_{\text{crit}}(B')$ suffices. A limit above the tipping point eventually lies above it. At equality, the exact pointwise comparison with the diverging threshold is indispensable. □

Two boundaries clarify the quantifiers. Mere upper boundedness of $B(t)$ is insufficient for the high-curvature exclusion: with $\ell(z) = z$, $\eta = 2$, $r = 0$, $S = p = d_0 = 1$, $B = a^{-2}$, the corner condition is $(1 + 1/a)^2 \le a$, true for $a \ge 4$. At logarithmic value, take $\ell(z) = z$, $r = 0$, $S = p = d_0 = 1$, so $B_{\text{tip}} = 1$. Along $B(t) = 1 - \exp(-t)$, the threshold is $a_{\text{crit}} = \exp(t)$. The paths $a(t) = \exp(2t)$ and $a(t) = \exp(t/2)$ give respectively inward and outward allocations for $t > 0$ despite having the same limiting budget.

A.7 A complete inward-basin characterization

Proof of proposition 14. For $z = B - px$, put $D = d_0 + a\ln(1+z) + rx$. This is positive, $D''(x) = -ap^2/(1+z)^2 < 0$, and therefore $(\ln D + Sx)'' = D''/D - (D'/D)^2 < 0$. The static optimizer is unique. It depends continuously on (B, a): rescale $x = (B/p)v$ to the compact fixed interval $v \in [0,1]$; continuity of the objective and uniqueness imply that every convergent subsequence of optimizing v converges to the unique limiting optimizer.

The derivative has the sign of

$$T(x; B, a) = r + Sd_0 + Srx + a\left[S\ln(1+z) - \frac{p}{1+z}\right]. \tag{73}$$

At fixed feasible x, its partial derivative in B is $a[S/(1+z) + p/(1+z)^2] > 0$. Increasing B also enlarges the feasible interval, so strict concavity implies that $x^*(B, a)$ is non-decreasing in B. At an interior optimum,

$$S\ln(1+z) - p/(1+z) = -(r + Sd_0 + Srx)/a < 0.$$

Thus increasing a makes the derivative at that old optimum strictly negative, and the new optimizer is strictly smaller. At $x = 0$, a non-positive derivative implies the same negative bracket, so a larger a preserves the zero optimum. At the other endpoint a new optimizer cannot exceed the feasible upper bound. These observations prove that x^* is non-increasing in a, and strictly decreasing whenever the old optimizer is interior and the new optimizer remains positive.

For $B < B_{\text{tip}}$, the zero derivative test is $a \geq a^{\ell}_{\text{crit}}(B)$. If this is met at some date, the budget stays constant, productivity rises strictly, and all subsequent choices remain zero. At or above the tipping budget, let $x_\dagger = (B - B_{\text{tip}})/p$. It is feasible and leaves local investment $z = B_{\text{tip}}$. The bracket in equation (73) vanishes there, while $r + Sd_0 + Srx_\dagger > 0$. Consequently $x^* > x_\dagger$, proving equation (43). Reaching equality with the tipping budget is followed by a strict crossing at the next date, after which the positive excess grows at least geometrically with multiplier $1 + g_X/p > 1$.

Call a path captured if it chooses zero in finite time, and crossed if it reaches or exceeds B_{tip} in finite time. A path that does neither has $x_t > 0$ and a non-decreasing budget bounded by B_{tip}. Hence $B_t \to B_\infty \leq B_{\text{tip}}$. If $B_\infty < B_{\text{tip}}$, the static productivity threshold is bounded along this path, whereas $a_t \to \infty$, forcing capture. Thus every unclassified path must converge to B_{tip} from below.

The captured set of initial productivities is nonempty: it contains $a_0 \geq a^{\ell}_{\text{crit}}(\bar{B})$. The crossed set is also nonempty. Choose a finite integer N such that $\bar{B}(1 + g_X/p)^N > B_{\text{tip}}$. Choose $a_0 > 0$ small enough that $a_0 \exp(g_a t) \leq Sd_0/p$ for $0 \leq t < N$. At the all-external endpoint the derivative numerator is $r + Sd_0 + SrB/p - ap \geq 0$, so the optimizer is all-external throughout those periods. The resulting budget crosses by period N.

Both sets are open as subsets of $(0, \infty)$. A captured path has $B < B_{\text{tip}}$, and one period after capture its corner inequality is strict because productivity has risen at unchanged budget. A crossed path is strictly above the tipping budget no later than one period after first reaching it. Continuity of every finite sequence of states and decisions preserves these strict finite-time certificates under sufficiently small changes of a_0.

The two sets are ordered. A larger a_0 gives smaller investment at a common budget; induction gives weakly smaller budgets at every date. If a smaller-a_0 path is captured, the larger one is captured no later than that date, since a^{ℓ}_{crit} is non-decreasing. Similarly, crossing of a larger-a_0 path implies crossing of a smaller one. Thus the captured set is an upper interval and the crossed set a lower interval.

There is at most one initial productivity in neither set. Suppose $a_0^- < a_0^+$ generated two such paths. Then $B_t^- \geq B_t^+$, and the gap is non-decreasing, because $x^*(B_t^-, a_t^-) \geq x^*(B_t^+, a_t^+)$. Both budgets would have to converge to B_{tip}, so their gap would have to remain identically zero. Yet for all sufficiently late dates, neither optimum is the all-external endpoint: at that endpoint its derivative numerator is $r + Sd_0 + SrB/p - ap$, which becomes negative for bounded B and $a \to \infty$. Neither optimum is zero by assumption. They are then interior, and the strict productivity comparison at a common budget forces a positive investment gap, a contradiction.

Finally, two disjoint nonempty open intervals cannot exhaust the connected set $(0, \infty)$. Their boundary contains a point in neither set, and the preceding argument makes that point unique. Denote it a_{sep}. The low-productivity construction makes it positive; openness of the captured set around $a^{\ell}_{\text{crit}}(\bar{B})$ makes it strictly smaller than that threshold. The boundary path has $B_t \uparrow B_{\text{tip}}$, and telescoping $B_{t+1} - B_t = g_X x_t$ gives its stated total investment.

For comparative statics, an increase of g_a increases every future productivity and weakly decreases investment and budgets by induction. An increase of g_X weakly increases the transition $B \mapsto B + g_X x^*(B, a)$, and an increase of $\bar{B}$ increases its initial state. Both therefore weakly increase all budgets. The ordered capture and crossing sets imply the asserted monotonicities of their unique separator. The comparison in $\bar{B}/B_{\text{tip}}$ holds with p, S, r, d_0 fixed, so that B_{tip} itself is fixed. □

Lemma 30 (No-further-investment certificate on a mobilization path)**.** *In the same one-channel logarithmic model, let a specified $B_{\mathrm{loc}}(t) > 0$ be non-decreasing, bounded by $\bar{B}$, and let $a_{T+k} = a_T \exp(g_a k)$ with $g_a > 0$. From state (T, X_T), zero investment at every date $T + k$ is optimal if and only if*

$$a_T \geq a_{\mathrm{stay}}(T, X_T) := \sup_{k \geq 0} \exp(-g_a k) a^{\ell}_{\mathrm{crit}}(B_{\mathrm{loc}}(T + k) + g_X X_T), \tag{74}$$

where the threshold is extended to infinity at and above B_{tip}. In particular, $\bar{B} + g_X X_T < B_{\mathrm{tip}}$ and $a_T \geq a^{\ell}_{\mathrm{crit}}(\bar{B} + g_X X_T)$ are sufficient. For the constant-$\bar{B}$ model starting at $X_0 = 0$, another sufficient capture certificate is the existence of an integer $T \geq 0$ such that

$$\bar{B}(1 + g_X/p)^T < B_{\mathrm{tip}}, \qquad a_0 \exp(g_a T) \geq a^{\ell}_{\mathrm{crit}}(\bar{B}(1 + g_X/p)^T). \tag{75}$$

Proof. If all future investments are zero, the stock is fixed at X_T and the static corner test must hold at every date. These tests are exactly the inequalities whose supremum appears in equation (74). Conversely, when they hold, induction gives the unique zero optimizer and unchanged stock at every date. The upper-budget certificate uses monotonicity of a^{ℓ}_{crit} and rising productivity. In the constant-ceiling model, $x_t \leq B_t/p$ implies $B_t \leq \bar{B}(1 + g_X/p)^t$. Under equation (75), the actual budget at date T is below the tipping budget and its threshold does not exceed the stated bound. Hence $x_T = 0$, after which capture is permanent. □

The cases excluded from the strict basin theorem have direct answers. If $g_X = 0$ and $\bar{B} < B_{\mathrm{tip}}$, the fixed budget and $g_a > 0$ ensure eventual capture for every $a_0 > 0$, but not necessarily immediate inwardness. If $g_a = 0$, $g_X > 0$, and the initial budget is below the tipping point, $a_0 \geq a^{\ell}_{\mathrm{crit}}(\bar{B})$ is captured immediately. Below that threshold, investment is initially positive and is non-decreasing as the budget grows; thus the tipping budget is crossed in finite time. If $\bar{B} \geq B_{\mathrm{tip}}$, every finite productivity is outward immediately. If $S = 0$, the finite residual tipping budget and the separator construction do not apply. At $B = 0$, the feasible set is a singleton and the divided corner formulas are unnecessary.

For the uniform bounded-budget extension at $\eta < 1$, concavity gives $\ell'(B) \geq \ell'(B_{\max}) > 0$ and $\ell(B) \leq \ell(B_{\max})$. The required price is therefore at most $[r_j + S_j(d_0 + a\ell(B_{\max}))^{\eta}]/[a\ell'(B_{\max})]$, which tends to zero. This proves the assertion even when budgets approach zero. It does not prove the analogous $\eta > 1$ statement without a positive lower budget bound.

A.8 Effective budgets with a finite productivity ceiling

Proof of proposition 12. Let $f(B) = a^{\ell}_{\mathrm{crit}}(B)$ for $0 < B < B_{\mathrm{tip}}$. Every denominator in equation (23) is positive there. The finite maximum of the resulting continuous ratios is continuous. The proof of proposition 11 establishes that f is non-decreasing and $f(B) \to \infty$ at B_{tip}. Thus its sublevel set $E = \{B \in (0, B_{\mathrm{tip}}) : f(B) \leq a_{\max}\}$ is downward closed. Divergence gives a neighbourhood below B_{tip} disjoint from E. If E is empty, the stated convention gives $B^* = 0$, and no positive budget is attainable under the ceiling. Otherwise its positive supremum lies strictly below B_{tip}. A sequence in E tending to that supremum, continuity, and downward closure prove $E = (0, B^*]$. If $f(B^*) < a_{\max}$, continuity would put a larger budget in E; hence $f(B^*) = a_{\max}$. The exact corner criterion now proves all attainability and endpoint claims. A larger ceiling enlarges E. For every fixed $b < B_{\mathrm{tip}}$, all ceilings at least $f(b)$ have $B^* \geq b$. Together with $B^* < B_{\mathrm{tip}}$ this proves convergence. The upper bound on B_{tip} follows from $\Gamma(B) \geq B$.

No strict monotonicity of f was used. For example, take $\ell(B) = B$, $d_0 = 1$, two unit prices, and $(r_1, S_1) = (0, 1)$, $(r_2, S_2) = (2, 0)$. Then $B_{\mathrm{tip}} = 1$ and $f(B) = \max\{(1 - B)^{-1}, 2\}$. A ceiling below two admits no positive budget; at two, $B^* = 1/2$ even though $f(0) = 2$. This plateau also warns against replacing existence by uniqueness of an inward optimizer. At $a = 2$, the second channel can exchange local and remote computation without changing total computation or residual value. Strict inequalities, not weak ceiling feasibility, justify the assertion about every optimizer in the revised path statement of proposition 11. □

A.9 Bounded productivity below unit value curvature

For any fixed $b_0 > 0$, concavity gives $\ell(B) \le \ell(b_0) + \ell'(b_0)(B - b_0)$ for $B \ge b_0$. The finite derivative therefore supplies a finite $K > 0$ with $\ell(B) \le KB$ eventually; differentiability at zero is unnecessary. A ceiling gives $\Omega \le a_{\max} KB$. For $0 < \eta < 1$, dividing equation (30) by B requires a quantity bounded above by $a_{\max} K$ to dominate a positive multiple of $B^{\eta/(1-\eta)}$, a contradiction. More generally the same calculation bounds any eventually inward productivity below by $K^{-1}(s/M)^{1/(1-\eta)} B^{\eta/(1-\eta)}$, proving $a(t) \to \infty$. At $\eta = 0$, equation (32) is exact. Replacing actual $a(t)$ by its upper bound yields only a necessary test. If its right-hand side is bounded below by a positive constant and $\ell'(B(t)) \to 0$, that test fails. Linear production, whose derivative does not vanish, can instead satisfy it with a fixed finite productivity.

A.10 The capped basin and its necessary qualification

Proof of proposition 15. The continuity and monotonicities of $x^*(B, a)$ proved in section A.7 do not depend on the productivity path. In case (i), put $\delta_x = x^*(\bar{B}, a_{\max}) > 0$. Since $B_t \ge \bar{B}$ and $a_t \le a_{\max}$, those monotonicities give $x_t \ge \delta_x$ until crossing and $B_t \ge \bar{B} + t g_X \delta_x$. Crossing therefore occurs no later than $\lceil (B_{\text{tip}} - \bar{B})/(g_X \delta_x) \rceil$. The runaway bound is a static inequality for every finite $a > 0$, so it continues to apply under either ceiling path.

For case (ii), the one-channel logarithmic threshold

$$f(B) = \frac{r + S d_0}{p/(1 + B) - S \ln(1 + B)}$$

is strictly increasing below B_{tip}. Write $b_* = B^*(a_{\max})$. Then $\bar{B} < b_* < B_{\text{tip}}$ and $f(b_*) = a_{\max}$. As before, call a path captured when some $x_t = 0$, and crossed when it reaches B_{tip}. Capture is permanent because productivity is non-decreasing and the budget then stops changing. Every captured path ends below b_*: a path with $a_0 < a_{\max}$ has $a_t < a_{\max}$ at all finite dates, while $a_0 = a_{\max}$ captures at $\bar{B} < b_*$.

An uncaptured, uncrossed path has positive investments and bounded increasing budgets, so $B_t \to B_\infty \le B_{\text{tip}}$ and $x_t \to 0$. Continuity and $a_t \to a_{\max}$ imply $x^*(B_\infty, a_{\max}) = 0$. The positive derivative at B_{tip} rules out that endpoint, and the corner test gives $B_\infty \le b_*$. At every finite date noncapture gives $a_t < f(B_t)$. Continuity below the tipping budget then gives $a_{\max} \le f(B_\infty)$, so $B_\infty \ge b_*$. Hence $B_\infty = b_*$, and telescoping yields $g_X \sum_{t \ge 0} x_t = b_* - \bar{B}$.

The captured initial values form an upper interval and crossed values a lower interval, by the policy monotonicities and pathwise ordering. The captured interval is nonempty since it contains $a_0 \ge f(\bar{B})$; the crossed interval is nonempty by the additional hypothesis. Both are relatively open in $(0, a_{\max})$: one period after capture the corner inequality is strict because $a_0 < a_{\max}$, while a crossing is strict at most one period after reaching B_{tip}. Finite-horizon continuity preserves these certificates. Connectedness supplies an initial value in neither interval.

There cannot be two such values. If $a_0^- < a_0^+$, their productivities remain strictly ordered at every finite date under equation (44). The corresponding budget gap is non-negative and non-decreasing. If both limits were b_*, this gap would be identically zero. Since $x_t \to 0$ and $B_t \ge \bar{B} > 0$, their positive optima are eventually interior; strict productivity monotonicity at the common budget then makes their investments different, a contradiction. The unique remaining value is a_{sep}. Openness of capture at $f(\bar{B}) < a_{\max}$ and the existence of a positive crossed value give $0 < a_{\text{sep}} < f(\bar{B})$. This proves the classification with no claim that every strictly ordered family supplies a nonempty crossed basin. □

Counterexample without a crossed path. Let $p = S = d_0 = g_X = 1$, $r = 0$, $\bar{B} = 1/10$, $a_{\max} = 10$, and $g_a = \ln 2$. For every $0 < a_0 \le 10$, the first budget satisfies $B_1 \le 1/5$, while

$a_1 = 5 + a_0/2 > 5$. Using $\ln(6/5) < 1/5$ gives

$$f(1/5) = \frac{1}{5/6 - \ln(6/5)} < \frac{30}{19} < 5.$$

Consequently all paths are captured by date one, although $a_{\max} > f(\bar{B})$ and the family preserves strict order at every finite date. Thus strict order alone does not establish a positive separator with both basins nonempty.

A hard cap changes the boundary classification. Keep $p = S = d_0 = g_X = 1$, $r = 0$, and $\bar{B} = 1/10$, but set $a_{\max} = f(1/5)$ and $a_t = \min(10^t a_0, a_{\max})$. For every $a_0 \in [a_{\max}/10, 1]$, the date-zero all-external endpoint is optimal because its derivative numerator is $1 - a_0 \geq 0$. Thus $B_1 = 1/5$; at date one productivity reaches the cap and the corner condition is an equality. Strict concavity makes $x_1 = 0$, and the path stays at $B_t = B^*(a_{\max}) = 1/5$ thereafter. For $0 < a_0 < a_{\max}/10$, the first budget is again $1/5$, but $x_1 > 0$ and $B_2 > B^*$. Once above B^*, the uniform lower bound $x^*(B_2, a_{\max}) > 0$ forces finite crossing of B_{tip}. The dividing initial value is still unique in this example, but its path is captured in finite time, and a whole interval of initial values shares the same terminal boundary budget. The unique *uncaptured* critical-path conclusion cannot be transferred to this coalescing family.

A.11 Shared technological change

Proof of proposition 7. The fixed B, d_0, S_j leave $H_j = (1 + B)\ln(1 + B)S_j$ unchanged. The transformed A_j is $(1 + B)(\lambda_r r_j + d_0 S_j)$. Divide these by p_j/κ and by $(p_j/\kappa)\lambda_a a$, respectively, and subtract the original remote-return term to obtain equation (29). No division by r_j, S_j, or A_j is involved, so zero slopes cause no exception. If $\lambda_r = \lambda_a$, the remote-computation component of ζ_j changes only by κ, whereas its residual component changes by κ/λ_a. In either case ξ_j changes by κ, independently of productivity. In particular, $\kappa\xi_j \geq 1$ in a nontrivial residual direction rules out inwardness at every finite transformed productivity. This does not assert that a particular physical technology realizes these multipliers. □

B Proofs for world valuation and residual substitution

B.1 World-number optimization

Proof of proposition 2. Eliminate W to obtain $C_{\text{tot}}^{\theta} c^{\alpha-\theta}$ on $[u, C_{\text{tot}}]$. If $\alpha > \theta$, its maximum is attained at $c = C_{\text{tot}}$ and $W = 1$. If $\alpha < \theta$, it is attained at $c = u$ and $W = C_{\text{tot}}/u$. Equality of the exponents makes every split tie. This proves equation (14), including $C_{\text{tot}} = u$. In either branch $V(u) = u^{\alpha}$ and $V(u(1 + C))/V(u) = (1 + C)^{v}$. Since $v > 0$, the stated affine normalization is $[(1+C)^{v} - 1]/v = F_{1-v}(C)$. An affine rescaling of the computational component requires the residual weights to be expressed in the same utility units; it is not neutral if those weights are held in unrelated units.

For logarithmic within-world welfare put $L = \ln(C_{\text{tot}}/u)$ and $v_c = \ln(c/u)$. The objective is $\exp[\theta(L - v_c)]v_c$ on $[0, L]$. At $\theta = 0$ it increases to L. At $\theta > 0$ its derivative has the sign of $1 - \theta v_c$, so $v_c^* = \min(L, 1/\theta)$. For $L \geq 1/\theta$ substitution gives the claimed power expression; below that boundary the maximum is L with one world. Consequently $\theta = 1$ gives linear value only above $\exp(1)u$ in this logarithmic case. No $\eta = 1$ value follows from the positive-α power model.

With integer W, eliminate c instead: the objective is $C_{\text{tot}}^{\alpha} W^{\theta-\alpha}$ for $1 \leq W \leq \lfloor C_{\text{tot}}/u \rfloor$. The same endpoint argument gives the stated integer formula when $\theta > \alpha$, and $W = 1$ when $\theta < \alpha$. Since $\lfloor t \rfloor/t \to 1$, the asymptotic exponent is unchanged. For regular counts W_d, the largest admissible count is between fixed positive multiples of C_{tot}/u for all sufficiently large

C_{tot}, because $W_{d+1} = bW_d + 1$. These bounds give the same logarithmic growth exponent; they do not give a smooth exact power function or a ratio converging to one. □

B.2 The optimized no-replication face

Proof of proposition 20. The feasible set is a compact simplex; the objective is continuous and concave, so a face optimizer exists. Differentiability and the positive budget give the usual multiplier conditions, including a multiplier λ for the budget and complementary multipliers for coordinates at zero. Equivalently, for each face coordinate v of unit resource price p_v, its marginal utility is at most $p_v\lambda$, with equality at positive coordinates. This can also be obtained directly by exchanges between any two positive-resource uses. At least one use is positive. The local derivative is positive, so the resulting λ is positive.

At a point with $x = y = 0$, the two missing marginal utilities are precisely the left sides of equation (52). If these are at most their prices times the same λ, then for every feasible point v the supporting-hyperplane bound is

$$U(v) - U(v^*) \le \nabla U(v^*) \cdot (v - v^*) \le 0.$$

The second inequality follows from the exhausted budget, equality in positive face coordinates, and the non-positive reduced derivatives at zero coordinates. Strict reduced derivatives for x, y make this bound strict whenever either is positive. Conversely, a full optimum satisfies the full multiplier conditions; restricting that multiplier to the face gives the stated existential condition. Thus the statement remains correct when several restricted optima or multipliers are possible. For $z^* > 0$, complementary slackness fixes $\lambda = F'(C^*)a\ell'(z^*)$. If $z^* = 0$, only $F'(C^*)a\ell'(0) \le \lambda$ is required, assuming the derivative is finite. The potentially non-differentiable endpoint is outside the present version of the proposition and can instead be treated with supergradients. If backups are unavailable, omit their coordinate and retain $R'(0)$ throughout. □

Proof of proposition 21 and the displayed reductions. For $w^* > 0$, $\epsilon_I I'(w^*) = p_m\lambda$; subtracting it from equation (52) proves equation (53) when no backup exists. For $b^* > 0$, subtract $p_b\lambda$ in the same manner. With both channels positive the industrial reduced derivative is $F'(C^*)r_h + (p_m + p_b - p_h)\lambda$, strictly positive when $p_h < p_m + p_b$. At equality and $r_h = 0$, non-decreasing concavity and $G_h'(0) = 0$ imply $G_h(y) = 0$ for $y \ge 0$. For $0 \le \epsilon \le \min(w^*, b^*)$, replace $(y, w, b_{\text{bk}}) = (0, w^*, b^*)$ by $(\epsilon, w^* - \epsilon, b^* - \epsilon)$. The budget, C, $x + y + b_{\text{bk}}$, and $x + y + w$ are unchanged, proving the tie statement. With a zero observation channel only $\epsilon_I I'(0) \le p_m\lambda$ is known; division by positive λ defines $c_I \le p_m$ and gives its exact shadow service cost. The backup argument is identical. Finally, at $q = 1$ the experiential marginal vanishes identically. Substitution proves equation (57); its sign and tie cases follow from non-negative returns and positive $F'(C^*), \lambda$. □

B.3 Expected value at risk

Proof of proposition 23. Let $k(x) = x/C(x)^\gamma$. Since $C(0) = \Omega > 0$, $k'(0) = \Omega^{-\gamma}$: the term differentiating the denominator is multiplied by $x = 0$. Also $C'(0) = r - pa\ell'(B)$ and $\varphi(0) = 0$. The complete product derivative is

$$U_R'(0) = (1 - \vartheta)F_\eta'(\Omega)[r - pa\ell'(B)] + \vartheta\varphi'(0)F_\eta(\Omega)\Omega^{-\gamma}.$$

A positive derivative improves along a feasible sufficiently small positive x, so an inward optimum requires this expression to be non-positive. Divide by $(1 - \vartheta)F_\eta'(\Omega) > 0$, rearrange, and use $a\ell'(B) = \Omega/\Gamma$ to obtain equation (55). For $\eta \ne 1$ exact algebra yields

$$K_\eta(\Omega) = \frac{(1 + \Omega) - (1 + \Omega)^\eta}{(1 - \eta)\Omega}.$$

For $0 \le \eta < 1$, $(1+\Omega)/\Omega \to 1$ and $(1+\Omega)^{\eta}/\Omega \to 0$, giving the claimed limit. In particular, for every $\varepsilon > 0$ there is Ω_ε such that $|K_\eta(\Omega) - (1-\eta)^{-1}| < \varepsilon$ for all $\Omega \ge \Omega_\varepsilon$. This supplies a rigorous two-sided asymptotic replacement even when Γ or the hazard varies; it is not an exact finite-output inequality with the limiting constant. The logarithmic expression follows directly from its definition, and its ratio to $\ln \Omega$ tends to one. For completeness, when $\eta > 1$, $K_\eta(\Omega) \sim \Omega^{\eta-1}/(\eta-1)$.

For seeds, a positive limiting K_η and bounded p force hazard odds times Γ to stay bounded. At a fixed budget, with $\Omega \to \infty$ and fixed r, the remote term tends to zero; taking limits and solving the resulting odds inequality proves equation (56). For mirrors at bounded positive budgets, $\Gamma r/\Omega \to 0$ and $\Gamma K_\eta(\Omega)/\Omega \to 0$ for $\eta \le 1$. With logarithmic value and seeds, the same exact inequality instead bounds hazard odds times $\Gamma \ln \Omega$. These consequences are necessary local comparisons unless additional global structure is supplied. At $\vartheta = 0$ the risk term vanishes. At $\vartheta = 1$, division by surviving marginal value is invalid; with $\varphi'(0) > 0$ and $F_\eta(\Omega) > 0$ the original derivative is positive and the inward corner fails. If $\varphi'(0) = 0$, this last first-order conclusion need not hold. The scaling specification itself requires $\Omega > 0$; no $0/0$ backup is defined at zero computation. □

B.4 Proof of the growing-budget separation

The restricted objective is strictly concave in (z, w, b) because the local log-composition and the two positive exponential utilities are strictly concave. It has a unique budget-exhausting optimum. If $z = 0$, the local marginal utility is B^2/d_0, whereas either positive service coordinate has marginal utility per resource at most a fixed constant. The budget multiplier conditions therefore rule out $z = 0$ for large B. With $z > 0$,

$$\lambda = \frac{B^2}{(1+z)[d_0 + B^2 \ln(1+z)]}.$$

We claim $z \to \infty$. Otherwise along a subsequence z is bounded, so this expression is bounded below by a positive constant. The service optimality conditions then imply

$$w = s_I \max\{0, \ln[J_I/(s_I p_m \lambda)]\}, \qquad b = s_R \max\{0, \ln[J_R/(s_R p_b \lambda)]\},$$

which bound both services on that subsequence. This contradicts $B = z + p_m w + p_b b \to \infty$. Hence $z \to \infty$ and $\lambda \sim [(1+z)\ln(1+z)]^{-1} \to 0$. Both services are eventually positive, and their formulas give $w + b = O(\ln z)$ and $B = z + O(\ln z)$. Consequently $z/B \to 1$. With $q = 1$, equation (52) reduces to the net-price conditions. Since $F'(C^*)/\lambda = (1+z)/B^2$, each required computational premium $r_j(1+z)/B^2$ tends to zero. The fixed positive margins $p_j - p_m - p_b$ make both conditions strict eventually, excluding replication in every optimizer. Finally, at the all-inward corner the service slopes J_I/s_I and J_R/s_R are fixed and positive, and bounded project prices contradict the growing logarithmic residual floor. This proves the claimed separation without assuming any resource-acquisition dynamics.

B.5 The no-replication construction under a finite ceiling

Keep the fixed parameters, logarithmic production and value, exponential service utilities, $q = 1$, and positive net prices of proposition 22, but allow any positive productivity path with $a \le a_{\max} < \infty$. The restricted face remains strictly concave and budget-exhausting. At least one of the expenditures $z, p_m w, p_b b$ is at least $B/3$. If it is z, its positive-coordinate multiplier condition yields

$$\lambda = \frac{a}{(1+z)[d_0 + a\ln(1+z)]} \le \frac{3a_{\max}}{d_0 B}.$$

If it is $p_m w$, put $u = w/s_I$ and use $ue^{-u} \le e^{-1}$: $\lambda p_m w = J_I u e^{-u} \le J_I/e$, whence $\lambda \le 3J_I/(eB)$. The backup case gives $3J_R/(eB)$. Consequently, with the fixed finite constant $K = 3\max\{a_{\max}/d_0, J_I/e, J_R/e\}$,

$$0 < \lambda \le K/B, \qquad \frac{F'(C^*)}{\lambda} \ge \frac{B}{K[d_0 + a_{\max}\ln(1+B)]} \longrightarrow \infty. \tag{76}$$

This uniform proof includes $z = 0$ and needs no positive lower bound on a. Since $\lambda \to 0$, each service's strictly positive marginal value at zero forces $w, b > 0$ eventually. The reduced test is then $r_j F'(C^*)/\lambda \le \Delta_j$, where $\Delta_j = p_j - p_m - p_b > 0$ is fixed. Every $r_j > 0$ eventually fails it. No no-replication face optimum is then a full optimum.

If $z > 0$, $\lambda = F'(C^*)a\ell'(z)$ gives the exact identity $F'(C^*)/\lambda = 1/[a\ell'(z)] = (1+z)/a$ and the exact condition $a\ell'(z) \ge r_j/\Delta_j$ at the optimized face point. The ceiling provides the necessary inequality with $a_{\max}$, not a sufficient condition for an arbitrary actual productivity or an unoptimized choice of z. If $r_j = 0$, that channel's reduced inequality is strict for $\Delta_j > 0$; if both replication slopes vanish, both channels pass. Non-positive net prices retain the separate sign and tie cases in proposition 21. At $z = 0$, use the multiplier test rather than the interior identity. For the pure-computation comparison in equation (80), the same logarithmic marginal production $a/(1+z)$ declines with growing local investment; the present calculation prices the optimized substitute services as well.

C Infinite-horizon verification and extensions

Proof of proposition 16. Let $A_- = \min(a_L, a_X/p) > 0$, $A_+ = \max(a_L, a_X/p)$, and define G_-, G_+ similarly. Every feasible share has $A(s) \in [A_-, A_+]$ and $G(s) \in [G_-, G_+]$. Consequently $B_0 G_-^t \le B_t \le B_0 G_+^t$ and

$$|\ln C_t| \le |\ln B_0| + t\max(|\ln G_-|, |\ln G_+|) + \max(|\ln A_-|, |\ln A_+|).$$

Since $\sum \beta^t(1+t) < \infty$, every objective is absolutely convergent, uniformly controlled at a fixed B_0. The same bound proves $\beta^T V(B_T) \to 0$ for the proposed V.

Substitute $V(B) = \ln B/(1-\beta) + K$ into the Bellman right side:

$$\max_s\{\ln[BA(s)] + \beta V(BG(s))\} = \frac{\ln B}{1-\beta} + \beta K + \max_s\{\ln A(s) + \tfrac{\beta}{1-\beta}\ln G(s)\}.$$

Thus equality requires $K = H_\beta^*/(1-\beta)^2$. A maximizing share exists by continuity on $[0,1]$. For any feasible policy, repeated use of the Bellman inequality gives

$$V(B_0) \ge \sum_{t=0}^{T-1} \beta^t \ln C_t + \beta^T V(B_T).$$

Letting $T \to \infty$ proves that V bounds its payoff. A stationary maximizing share gives equality at every step and hence attains V. This is a verification theorem, not just a proposed functional form. Any other Bellman solution with growth bounded by a constant times $1 + |\ln B|$ has the same vanishing tails and the same telescoping comparison, so it agrees with this value.

The second derivative of H_β is

$$-(1-\beta)\frac{(a_X/p - a_L)^2}{A(s)^2} - \beta\frac{(g_X/p - g_L)^2}{G(s)^2} \le 0.$$

Therefore the inward endpoint maximizes it exactly when $H'_\beta(0) = (1-\beta)[(a_X/p)/a_L - 1] + \beta[(g_X/p)/g_L - 1] \le 0$. If at least one of the two coefficients differs, strict concavity gives a unique maximizing share. If both are equal, all shares tie and the inward policy is among the optimal policies. Endpoints and ties are thus included. At $\beta = 0$ the immediate problem is recovered separately; at $\beta = 1$ the discounted theorem is not asserted. Allowing zero production or growth endpoints would destroy the uniform logarithmic bounds and is not covered. □

Proof of proposition 19 and finiteness statements. Under $\beta G_+ < 1$, $0 \le Sx_t \le SB_0 G_+^t/p$, so the residual series converges uniformly over policies. The preceding logarithmic bound controls the other term, proving finite values. Under all-external allocation, $x_t = B_0(g_X/p)^t/p$; hence the residual series diverges when $\beta g_X/p \ge 1$. Its logarithmic component is finite and cannot cancel this divergence.

The all-inward policy has zero residual reward and continuation $V_L(B) = \ln B/(1-\beta) + K_L$, where $K_L = \ln a_L/(1-\beta) + \beta \ln g_L/(1-\beta)^2$. A deviation x now followed by that policy has value

$$J_B(x) = \ln[a_L(B - px) + a_X x] + Sx + \beta V_L[g_L(B - px) + g_X x].$$

At zero its derivative is

$$J_B'(0) = S + \frac{p}{B}\left[\frac{a_X/p}{a_L} - 1 + \frac{\beta}{1-\beta}\left(\frac{g_X/p}{g_L} - 1\right)\right].$$

If the all-inward policy is globally optimal it must defeat this feasible one-step deviation, so $J_B'(0) \le 0$. Rearrangement gives equation (49). The right side is fixed while SB/p diverges. This proves the obstruction without claiming that the displayed condition suffices for a general mixed policy. □

One-step ceiling implication in equation (50). For $x \ge 0$,

$$\min(\bar{B}, g_L(B - px)) \ge \min(\bar{B}, g_L B) - pg_L x.$$

Thus $g_X \ge pg_L$ makes next-period resources at least their inward value. An increasing continuation value cannot penalize this deviation. In the logarithmic current-utility model, proposition 11 makes its current right derivative strictly positive at every finite productivity when $B \ge B_{\text{tip}}$. A sufficiently small positive x therefore raises current payoff and does not reduce future payoff. This excludes an optimal inward first action. The same inequality covers the kink $g_L B = \bar{B}$ without differentiating through it. A finite value and an increasing continuation are maintained assumptions here; no homogeneous formula is substituted after adding the ceiling. □

D Latency bounds and their domain

For proposition 24, the exact required price is

$$P_j(t) = \exp[-\mathcal{D}_j(t)]\left\{\frac{(1+B)(r_{j0} + d_0 S_{j0})}{a} + \Gamma(B) S_{j0}\right\}.$$

Put $H(B) = (1+B)[1 + \ln(1+B)]$. Both $1+B$ and $\Gamma(B)$ are bounded above by $H(B)$. Thus $P_j(t) \le \exp[-\mathcal{D}_{\min}(t)] H(B(t)) K_j$, where $K_j = (r_{j0} + d_0 S_{j0})/a_{\min} + S_{j0}$ is finite. Under equation (59) the right side tends to zero, and hence is strictly below $p_{\min}$ eventually. Finitely many directions permit one common eventual time. If $B \ge B_{\min} > 0$, then $H(B)/\Gamma(B) = 1 + 1/\ln(1+B)$ is bounded, proving the alternative. For non-decreasing ω, its integral over an interval of length T_j^{lag} is at least $\omega(t) T_j^{\text{lag}}$. Finally both $1+B$ and $\Gamma(B)$ increase in B, so bounding them at $\bar{B}$ and a below by $a_{\min}$ proves equation (60).

Neither omitted condition can be dropped from the proposed version. Take $B(t) = \exp(-t)$, $D(t) = 0$, $a = p = 1$, $r = 2$, $S = 0$, $d_0 = 1$. Then $D - \ln \Gamma(B) \to \infty$, but the required price $2(1+B)$ exceeds one for every t. Alternatively let $B = 1$, $a = 1$, $r = 1$, $S = 0$, $D = t$, and $p(t) = \exp(-2t)$. Discounting beats any fixed budget factor, but the required price $2\exp(-t)$ exceeds the actual price. These examples show why an upper price bound cannot substitute for the lower price bound in this sufficient condition. If all base returns vanish, every direction already passes without a latency assumption. Equality at a finite threshold establishes only a weak corner condition, not uniqueness.

E Population likelihoods and strategic interaction

For a fixed list, independence multiplies individual zero-arrival probabilities $\exp(-\mu_{A,i})$, yielding

$$P_0^{\text{fixed}} = \exp(-\sum_i \mu_{A,i}). \tag{77}$$

For finite origin intensity $\bar{N}_A$, condition on a Poisson number K of origins and independent marks drawn from the normalized intensity. With $m = \mathbb{E}\exp(-\mu_A)$,

$$P_0 = \sum_{k=0}^{\infty} e^{-\bar{N}_A} \frac{\bar{N}_A^k}{k!} m^k = \exp[-\bar{N}_A(1-m)],$$

proving equation (65). Restriction to increasing finite-intensity regions and continuity of probabilities proves the corresponding sigma-finite formulation whenever needed, allowing a zero limit. This is the generating-functional identity of Last and Penrose (2018).

For proposition 27, monotonicity of $1-e^{-u}$ implies $\mathbb{E}(1-e^{-\mu_A}) \geq \pi_{\text{out}}(1-e^{-\mu_{\text{out}}})$. The hypothesis $P_0 \geq P_*$ bounds this expectation by $\mathcal{L}_*/\bar{N}_A$, proving the first inequality. Its inward contribution is non-negative and bounded by the same total, giving the leakage inequality. Two constant marks evaluate the expectation as $f_{\text{in}} + \pi_{\text{out}}(f_{\text{out}} - f_{\text{in}})$. If $\mathcal{L}_*/\bar{N}_A < f_{\text{in}}$ no mixture works; if it is at least f_{out} every mixture works. Otherwise the exact upper fraction is $(\mathcal{L}_*/\bar{N}_A - f_{\text{in}})/(f_{\text{out}} - f_{\text{in}})$. At equal means the fraction cancels and only the common mean matters. Since $0 \leq 1-e^{-\mu_A} \leq 1$, $P_0 \geq e^{-\bar{N}_A}$ proves the universal rarity sufficient condition. A lower mean bound alone cannot improve it: take $\bar{N}_A = 100$, $P_* = .05$, and lower bound $\mu_{\text{out}} = .01$, but actual means twenty for every origin. The proposed weaker rarity expression is below $\mathcal{L}_*$, while the actual null probability is near e^{-100}. If $\bar{N}_A = 0$, the null probability is one without restrictions. If $\mu_{\text{out}} = 0$, division in equation (66) is invalid and no bound follows from that zero lower bound. Fractions zero and one and equality $P_0 = P_*$ are included by the weak inequalities. Inward leakage averages the full function of the random mark, not that function at its mean.

E.1 Proof of the preemption proposition

Fix rivals in equation (68). The own-action objective is strictly concave, so its unique maximizer satisfies exactly the three conditions in equation (69). Simultaneous satisfaction is the definition of Nash equilibrium. The continuous function $\Phi(x) = \sum_i f(x_i) + \iota \sum_{i<k} x_i x_k$ attains a maximum on the compact cube. A unilateral change changes Φ by precisely the change in that player's payoff, so such a maximum is an equilibrium. Substituting all-zero and all-cap profiles proves their conditions; substituting a common interior action gives the stated scalar equation. This characterizes asymmetric profiles too through the coordinate conditions; it does not assert that every equilibrium is symmetric.

For binary choices, the payoff difference between cap and zero is $\bar{X}[d_{\text{bin}} + \iota X_{-i}]$. Its sign proves the two endpoint criteria. With one maintained seed and all others initially zero, the remaining players initially see difference proportional to $d_{\text{bin}} + \iota\bar{X}$. If positive, each first updating player adopts, and the incentive never decreases as others adopt. Under a fair asynchronous update schedule, all adopt in finitely many individual updates. If non-positive, none adopts with inward tie-breaking, so the seed does not trigger a cascade. This proves the claimed iff under the specified update and tie rules.

For the continuous counterexample, best response is $\max\{0, (2x_{-i}-1)/10\}$ on $[0,1]$. At an equilibrium both actions are at most 0.1, which forces both responses to zero. Hence the sole equilibrium is zero despite the positive origin derivative against a maximal rival. The binary five-player counterexample follows by direct substitution. These calculations refute an unconditional one-defector cascade; they do not exclude cascades in games satisfying the stated

strict condition. The thermal identity in equation (71) is algebra, not an additional behavioral theorem.

F Numerical specification

All random tests use NumPy generators with fixed seeds recorded in each script. The present environment is Python 3.13.5, NumPy 2.3.5, and SciPy 1.17.0. The reference design is the archived 14-dimensional, 1,024-row scrambled Sobol array, generated with `seed=20260905`, not the newer `rng` keyword. No regenerated array replaces it. Its source file SHA-256 is

```
ce45ecd0435f2abf834445957db369d1d4a5bc28e9c4e7b686561df50a14194b
```

The range transformation, scenario table, and archived measure sensitivity are in the supplement. Every row of the new no-replication classification uses those same parameter values, with the original instantaneous prices, $q = .99$, $\epsilon_R = .01$, and $\epsilon_I = .05$.

Equation checks use explicit absolute and relative tolerances printed in the scripts, generally 10^{-8} or tighter after scale normalization. Nonconcave objectives are checked with dense scalar candidate searches and local refinement; passing these checks is not a proof of global optimality. The exact endpoint theorem is proved separately. Concave full-objective comparisons use multiple starting points where specified and reduced-gradient certificates; the frozen no-replication ensemble additionally uses a scalar root for the observation optimum. The infinite-horizon check maximizes each finite-horizon Bellman update independently; its largest absolute value error in 180 tests was 8.293×10^{-11}. The full mathematical transversality argument does not depend on that finite truncation.

New verification covers service-boundary and bundle cases, exact hazard product derivatives, continuous and integer world counts, the latency integral, marked-origin mixtures, continuous and binary strategic games, and thermal root solutions. The new growing-budget no-replication construction uses a restricted-face resource root and tests the full replication inequalities. Results are machine-readable JSON plus literal stdout, accompanied by a runner which fails on any failed assertion. No script imports the production implementation of another new validation script. Shared NumPy/SciPy routines do not constitute independent peer review. The README maps the validation scripts to their mathematical targets and distinguishes the v8 checks from the archived reruns.

G Notation and scope dictionary

Symbol	Meaning
B, z, p_j	Budget and local spending in fixed resource-reference units; external price per unit project capacity.
$b_{\text{ref}}, c_{\text{ref}}$	Physical resource and computational-service reference units.
x, y, w, b_{bk}	New resident, industrial, observation, and dormant-backup capacities.
n, h, m	Nested cumulative residence, industry, and observation indices.
$C, C_0, a, \ell, G_j, r_j$	Useful computation, baseline, local productivity, local production shape, external production and its initial slope.
F_η, η, η_k	Shifted isoelastic value and curvature; service-specific efficiency has a subscript.
$q, \sigma(q), E, R, I$	Experiential substitution and residual experience, resilience, information functions.
S_j, H_j, A_j	Residual corner slope and its logarithmic floor/productivity coefficients.
Ω, Γ, d_0	Local corner output $a\ell(B)$, production ratio ℓ/ℓ', and $1 + C_0$.

Symbol	Meaning
λ_L, D_j, λ	Original local and external marginal values; restricted-face budget multiplier.
$\bar{B}, B_{\mathrm{tip}}, a_{\mathrm{sep}}, g_a, g_X, X_t$	Local ceiling, fixed-residual tipping budget, basin separator, productivity growth, acquisition yield, and inherited external stock.
$\rho, \kappa, \xi_j, \zeta_j$	All-inward robustness ratio, uniform price multiplier, and logarithmic phase coordinates.
κ_j	External production curvature; not the common price multiplier.
b, d, W_d, Q, χ_T	Branching, depth, world count, support capacity ratio, and maximum demand-to-capacity ratio.
$u, c, C_{\mathrm{tot}}, \theta, \alpha$	Minimum and actual per-world resources, total world resources, population weight, and per-world welfare elasticity.
$\vartheta, \varphi, \gamma$	Horizon catastrophe probability, mitigation probability, and backup scale exponent.
$\beta, a_L, a_X, g_L, g_X$	Objective-period discount, linear computation coefficients, and resource multipliers in the dynamic benchmark.
$T_j^{\mathrm{lag}}, \omega, \delta, \mathcal{D}_j(t)$	Objective delay, subjective clock, subjective discount rate, and integrated delay exponent.
$\mu, \mu_A, q_R, R_{\mathrm{rogue}}$	Successful-front mean, arrival mean, extinction probability, and offspring mean.
$\nu, \bar{N}_A, \pi_{\mathrm{out}}, \pi_j$	Origin intensity, expected relevant origin count, outward fraction, and survey selection probability.
$N_i^{\mathrm{act}}, \lambda_i, \iota$	Physically outward-capable nodes, per-node initiation rate, and preemption sensitivity.
$\Lambda_j, \mathcal{P}_j, \tau$	Detection mean, source-state probability, and source lifetime.
s_E, s_R, s_I	Benefit saturation scales; stored archive column names remain e, r, i.
$L_*, \mathcal{L}_*$	Host luminosity and null-likelihood exposure allowance $-\ln P_*$.
$L_\odot^{\mathrm{N}}, \sigma_{\mathrm{SB}}, M_{\mathrm{rad}}, \Sigma_{\mathrm{rad}}$	Nominal solar luminosity, Stefan–Boltzmann constant, radiator mass, and areal mass.

Physical scale is not fixed by normalized parameter endpoints alone. The model's resource horizon and the project-capacity units must accompany any numerical interpretation. Original archive variable names are retained in data files for byte-level reproducibility, but their displayed notation has been disambiguated.

References

Fernando Alvarez and Nancy L. Stokey. Dynamic programming with homogeneous functions. *Journal of Economic Theory*, 82(1):167–189, 1998. doi: 10.1006/jeth.1998.2431.

Stuart Armstrong and Anders Sandberg. Eternity in six hours: Intergalactic spreading of intelligent life and sharpening the Fermi paradox. *Acta Astronautica*, 89:1–13, 2013. doi: 10.1016/j.actaastro.2013.04.002.

Charles H. Bennett. Logical Reversibility of Computation. *IBM Journal of Research and Development*, 17(6):525–532, 1973. doi: 10.1147/rd.176.0525.

Charles H. Bennett, Robin Hanson, and C. Jess Riedel. Comment on 'The Aestivation Hypothesis for Resolving Fermi's Paradox'. *Foundations of Physics*, 49:820–829, 2019. doi: 10.1007/s10701-019-00289-5. arXiv:1902.06730.

Nick Bostrom. Astronomical Waste: The Opportunity Cost of Delayed Technological Development. *Utilitas*, 15(3):308–314, 2003. doi: 10.1017/S0953820800004076.

Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004. doi: 10.1017/CBO9780511804441. URL `https://web.stanford.edu/~boyd/cvxbook/`.

Robert J. Bradbury. *Matrioshka Brains*. Non-peer-reviewed working manuscript, created 21 July 1999; mirrored version last modified 16 August 2004. `https://ludios.org/thirdparty/MatrioshkaBrainsPaper.html`.

G. D. Brin. The "Great Silence": the controversy concerning extraterrestrial intelligent life. *Quarterly Journal of the Royal Astronomical Society*, 24(3):283–309, 1983.

Jonathan Carroll-Nellenback, Adam Frank, Jason T. Wright, and Caleb Scharf. The Fermi Paradox and the Aurora Effect: Exo-civilization Settlement, Expansion, and Steady States. *The Astronomical Journal*, 158(3):117, 2019. doi: 10.3847/1538-3881/ab31a3.

Milan M. Ćirković and Robert J. Bradbury. Galactic Gradients, Postbiological Evolution and the Apparent Failure of SETI. *New Astronomy*, 11(8):628–639, 2006. doi: 10.1016/j.newast.2006.04.003.

Steven J. Dick. Cultural Evolution, the Postbiological Universe and SETI. *International Journal of Astrobiology*, 2(1):65–74, 2003. doi: 10.1017/S147355040300137X.

Robin Hanson. *Burning the Cosmic Commons: Evolutionary Strategies for Interstellar Colonization*. Working paper, 1 July 1998. `https://mason.gmu.edu/~rhanson/filluniv.pdf`.

Jacob D. Haqq-Misra and Seth D. Baum. The Sustainability Solution to the Fermi Paradox. *Journal of the British Interplanetary Society*, 62:47–51, 2009. doi: 10.48550/arXiv.0906.0568. arXiv:0906.0568.

Theodore E. Harris. *The Theory of Branching Processes*. Springer, Berlin, 1963.

Michael H. Hart. An Explanation for the Absence of Extraterrestrials on Earth. *Quarterly Journal of the Royal Astronomical Society*, 16:128–135, 1975.

International Astronomical Union. Resolution B3: Recommended nominal conversion constants for selected solar and planetary properties. Adopted 13 August 2015; explanatory document by E. E. Mamajek and colleagues, arXiv:1510.07674, 2015. `https://arxiv.org/abs/1510.07674`.

Jet Propulsion Laboratory, Solar System Dynamics. *Planetary Physical Parameters*. NASA/JPL institutional data table, accessed 13 October 2026. `https://ssd.jpl.nasa.gov/planets/phys_par.html`.

Rolf Landauer. Irreversibility and Heat Generation in the Computing Process. *IBM Journal of Research and Development*, 5(3):183–191, 1961. doi: 10.1147/rd.53.0183.

Günter Last and Mathew Penrose. *Lectures on the Poisson Process*. Cambridge University Press, 2018. doi: 10.1017/9781316104477.

William I. Newman and Carl Sagan. Galactic Civilizations: Population Dynamics and Interstellar Diffusion. *Icarus*, 46(3):293–327, 1981. doi: 10.1016/0019-1035(81)90135-4.

National Institute of Standards and Technology. *CODATA Internationally Recommended 2022 Values of the Fundamental Physical Constants*. Institutional publication record, 8 May 2024, updated 29 August 2024; authors Peter J. Mohr, Eite Tiesinga, David B. Newell, and Barry N. Taylor. `https://www.nist.gov/publications/codata-internationally-recommended-2022-values-fundamental-physical-constants`.

Anders Sandberg, Stuart Armstrong, and Milan M. Ćirković. That Is Not Dead Which Can Eternal Lie: The Aestivation Hypothesis for Resolving Fermi's Paradox, 2017. arXiv:1705.03394.

John M. Smart. The Transcension Hypothesis: Sufficiently Advanced Civilizations Invariably Leave Our Universe, and Implications for METI and SETI. *Acta Astronautica*, 78:55–68, 2012. doi: 10.1016/j.actaastro.2011.11.006.

Matías Suazo, Erik Zackrisson, Priyatam K. Mahto, Fabian Lundell, Carl Nettelblad, Andreas J. Korn, Jason T. Wright, and Suman Majumdar. Project Hephaistos – II. Dyson Sphere Candidates from Gaia DR3, 2MASS, and WISE. *Monthly Notices of the Royal Astronomical Society*, 531(1):695–707, 2024. doi: 10.1093/mnras/stae1186. arXiv:2405.02927.

Frank J. Tipler. Extraterrestrial Intelligent Beings Do Not Exist. *Quarterly Journal of the Royal Astronomical Society*, 21:267–281, 1980.

Michael L. Wong and Stuart Bartlett. Asymptotic Burnout and Homeostatic Awakening: A Possible Solution to the Fermi Paradox? *Journal of the Royal Society Interface*, 19(190): 20220029, 2022. doi: 10.1098/rsif.2022.0029.

Jason T. Wright, Roger L. Griffith, Steinn Sigurdsson, Matthew S. Povich, and Brendan Mullan. The $\hat{\mathrm{G}}$ Infrared Search for Extraterrestrial Civilizations with Large Energy Supplies. II. Framework, Strategy, and First Result. *The Astrophysical Journal*, 792(1):27, 2014. doi: 10.1088/0004-637X/792/1/27.

Supplementary Material

S1 Retained workload and budget illustrations

The numerical examples below are retained from v7.3. Their figures and frozen arrays are unchanged; only their placement and notation have been updated. Four controlled examples use $b = 2$, $d(t) = 2 + \lfloor t/10 \rfloor$ on $0 \le t \le 40$, and local capacities $(32, 256, 24)$ for update work, stored-state memory, and heat rejection. All $u_k = 1$. Processing and thermal efficiency either grow as $e^{0.12t}$, grow as $e^{0.01t}$, or grow as $e^{0.12t}$ up to a ceiling of four; memory efficiency remains one. A fourth example activates at most eight worlds at a time without efficiency improvement. Every example stores all W_d states. These are normalized accounting constructions, not device specifications.

The matched-efficiency example remains feasible while the full hierarchy grows from seven to 127 worlds. Demand outpacing efficiency first violates a constraint at $t = 20$, and the efficiency-ceiling example first violates one at $t = 40$. Scheduling remains feasible on the tested interval, but only eight worlds are active at once. The maximum ratios are respectively 0.4961, 3.5471, 1.3229, and 0.4961. All changes in depth occur at known times, so interval maxima can be checked at the jumps rather than inferred from a sampled time grid alone. Figure S1 illustrates the distinction between technology-supported expansion of internal activity and merely postponing a resource limit. Extending the matched-efficiency path with the same fixed memory efficiency and capacity gives $W = 255$ and a memory ratio $255/256 = 0.99609375$ at $t = 50$, but $W = 511$ and $511/256 = 1.99609375$ at $t = 60$. The latter is its first memory violation. The example therefore supports the stated finite horizon, not indefinite local sufficiency.

S1.1 A joint finite-interval witness

The support and allocation examples need not be merely disconnected illustrations. Couple the matched-efficiency schedule in section 2 to $B = 10$, $C_0 = 0$, $a(t) = W_{d(t)}/\ln(11)$, $r_n = r_h = 0.1$, and all three prices equal to one. Set $q = 0.99$, $\epsilon_R = \epsilon_I = 0.001$, with unit initial slopes for E, R, I. At $z = B$, useful computation is exactly $C_L = W_{d(t)}$, the workload charged in the support example. The three outward-to-local marginal-return ratios are

$$11\ln(11)\left[S_j + \frac{r_j + S_j}{W_{d(t)}}\right], \qquad (S_n, S_h, S_m) = (0.012, 0.002, 0.001). \tag{78}$$

They decrease with world count and are already below one at seven worlds. Thus this same finite family satisfies both the three-service feasibility test and strict all-inward optimality with positive residual external values. As the hierarchy grows from seven to 127 worlds, attained logarithmic value rises from $\ln 8$ to $\ln 128$. The budget is fixed at $B = 10$ and productivity is deliberately tuned to $a(t) = W_{d(t)}/\ln 11$. This is a consistency witness, not evidence for the plausibility of that productivity path, an engineering implementation, or an infinite-horizon optimum. Persistent memory is still charged to every world; future resource effects are considered separately in section 4.5.

S1.2 Budget growth and shared technological progress

With perfect experiential substitution and no residual external value, $q = 1$ and $\epsilon_R = \epsilon_I = 0$. For *any* strictly increasing F, maximizing value is equivalent to maximizing C. Let

$$R_* = \max\{r_n/p_n, r_h/p_h\}. \tag{79}$$

When $R_* > 0$, the solution allocates

$$z^* = \min\{B, \max(a/R_* - 1, 0)\}, \qquad B_{\text{crit}} = a/R_* - 1. \tag{80}$$

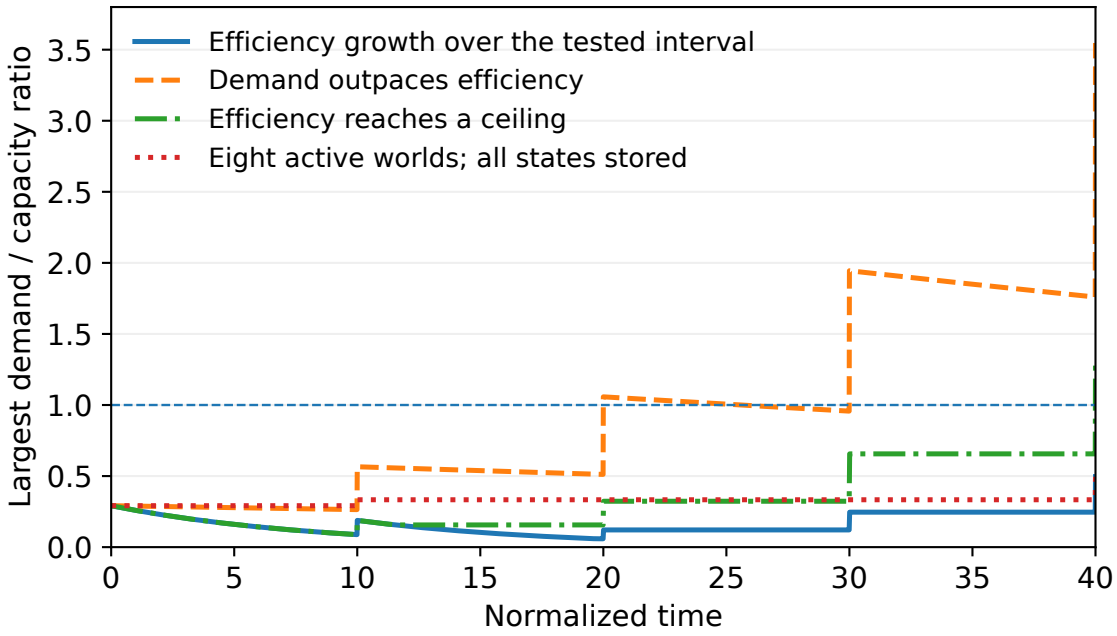


Figure S1: Finite-horizon local support constructions. The largest service-demand ratio must remain at or below one. Processing and heat are charged to active worlds; persistent memory is charged to all represented states, including inactive ones. Growth in depth is stepwise. No curve is extrapolated to unlimited time, and no physical implementation is validated.

Any remaining budget goes to a best-return computational external channel. The all-inward condition is $a/(1+B) \geq R_*$. If $R_* = 0$, all investment is local. This exposes a limitation of a fixed-budget productivity path: with fixed a and $R_* > 0$, a sufficiently large budget restores external investment even without authenticity, information, or resilience premiums. The civilization does not stop preferring more value when it can generate virtual worlds.

If local and remote computational coefficients all receive the same multiplicative improvement, the pure-computation condition is unchanged. Progress that disproportionately raises intensive return can support inward allocation; progress that improves everything equally need not. Figure S2 shows this budget effect for $R_* = 1$. Persistent inward allocation along a development path requires the relevant inequalities to hold along that path, not merely at one chosen value of B.

S2 Paired sensitivity design and numerical specification

S2.1 Paired sensitivity design

To distinguish deployment costs from access delays, use the normalized family

$$p_j = p_{j0}\left(f + \frac{1}{s_t} + \frac{0.2}{s_c}\right), \qquad r_j = r_{j0} m_r \exp(-0.1/s_t), \qquad a = a_0 m_a. \tag{81}$$

Here $f > 0$ is the residual price factor; $s_t, s_c > 0$ index effective material and information access speeds relative to c. At instantaneous access, $p_j = p_{j0} f$ and $r_j = r_{j0} m_r$. The coefficients 0.2 and 0.1 define scenarios, not calibrated physical laws. Speed sweeps also remove the stipulated attenuation of remote returns, so they are *not* pure price experiments. The fixed-access sweeps in f isolate the price effect.

Faster-than-light access is only a counterfactual limiting case of this map, not a physical mechanism established here. Instantaneous arrival still leaves the positive deployment price $p_{j0} f$.

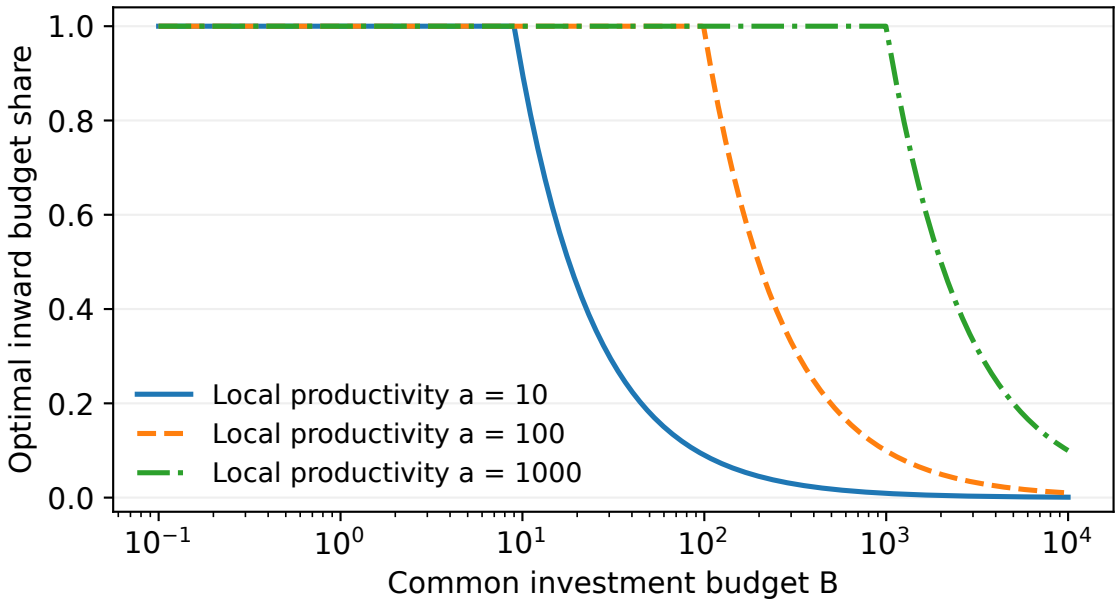


Figure S2: Exact inward budget share with perfect experiential substitution, zero residual value, and $R_* = 1$. The transition occurs at $B = a - 1$. Greater local productivity moves the boundary, but budget growth can eventually favor additional external capacity. Any strictly increasing computational value function gives the same allocation in this pure-computation case.

Exactly zero external prices are excluded: they can destroy compactness of the feasible set and, with positive external computational returns and unbounded increasing value, need not admit a finite optimizer.

The reference design contains 1,024 scrambled Sobol points in 14 dimensions, paired across every scenario. The exact library, generator call and frozen inputs are specified in section F. Marginals are log-uniform over the stated normalized ranges. The concave residual benefits are

$$E(x) = A_E(1 - \exp(-x/s_E)), \quad R(u) = A_R(1 - \exp(-u/s_R)), \quad I(v) = A_I(1 - \exp(-v/s_I)). \tag{82}$$

The reference uses $(q, \epsilon_R, \epsilon_I, f) = (0.99, 0.01, 0.05, 1)$ and $m_a = m_r = 1$. Table S2 retains the full paired counts.

At instantaneous access, the reference has 219 inward corners; a uniform 10% price cut leaves 207. Only 364 reference inputs admit a finite productivity threshold with the same residual slopes; the other 660 violate a residual-floor condition. Perfect experiential substitution with zero residual weights yields 527 inward corners, while 497 still prefer external computational capacity. Raising only local productivity tenfold gives 314 inward corners; raising local and remote productivity together gives 222. At $f = 0.01$ and $f = 10^{-4}$, the count is zero. These are exact classifications of the archived inputs, not estimates of civilizational prevalence.

Eight additional scrambles of 4,096 points each give instantaneous inward fractions of 20.31–21.07% under the same log-uniform marginals. Linear-uniform marginals over the same endpoints give 10.99–11.82%. These 65,536 classifications use the exact boundary. Their variation describes sensitivity to the parameter-space measure, not observational uncertainty about the Galaxy.

S2.2 Frozen design and retained paired results

The canonical unit points and transformed parameters are the arrays `sobol_vectors` and `parameters` in `data/allocation/design.npz`. The generating environment records Python 3.13.5, NumPy 2.3.5 and SciPy 1.17.0. The exact call is

```
from scipy.stats import qmc
u = qmc.Sobol(d=14, scramble=True,
              seed=20260905).random_base2(m=10)
```

The default `bits=None` resolves to 30 bits; optimization is `None`. No points are skipped or thinned. For the ordered endpoint vectors in table S1, the transformation is

```
lo = np.log(lower_endpoints)
hi = np.log(upper_endpoints)
parameters = np.exp(lo + u * (hi - lo))
```

Every scenario uses the same transformed rows.

Re-execution in SciPy 1.17.0 reproduces the archived unit points bit for bit. Floating-point exponentiation changes some transformed entries by at most a relative 2.22×10^{-16} in the present environment; all classifications agree. Evaluation of the frozen arrays reproduces 219 instantaneous-reference inward corners and 527 under perfect substitution with zero residual weights, as well as all other counts in table S2.

With `seed=20260905`, the archived unit points are reproduced bit for bit in the historical checks under SciPy 1.17.0 and 1.18.1; `rng=20260905` yields a different scramble, with 208 instantaneous and 530 perfect-substitution inward corners. The frozen arrays, not the seed alone, define the reference experiment. The frozen package and the accompanying integrity manifest record the generator and checksums. The present rerun uses SciPy 1.17.0; the 1.18.1 result is retained archival evidence.

The parameter order is $(B, C_0, a, r_n, r_h, A_E, A_R, A_I, s_E, s_R, s_I, p_{n0}, p_{h0}, p_{m0})$. All base marginal ranges are log-uniform in the reference experiment. The stored columns named e, r, and i are written s_E, s_R, s_I here; their values and ordering are unchanged.

Table S1: Normalized ranges in the 14-dimensional reference design.

Quantity	Range	Meaning
B	1–100	common investment budget
C_0	0.1–10	existing useful computational service
a	1–1000	intensive computational coefficient
r_n, r_h	0.1–10	external computational returns
A_E, A_R, A_I	0.1–10	benefit amplitudes
s_E, s_R	1–30	experiential and resilience scales
s_I	1–100	information scale
p_{n0}	1–10	resident unit price
p_{h0}	0.1–3	industry-only unit price
p_{m0}	0.01–0.3	observation-only unit price

The sublight scenario has $(s_t, s_c) = (0.01, 1)$, the communication-only variant $(0.01, \infty)$, the light-speed case $(1, 1)$, and combined faster-access cases $(10, 10)$, $(1000, 1000)$, and (∞, ∞). The no-substitution variant sets $q = 0$. Large residual values set $\epsilon_R = \epsilon_I = 1$, either retaining $q = 0.99$ or additionally setting $q = 0$. The two tenfold productivity variants set $(m_a, m_r) = (10, 1)$ and $(10, 10)$. The perfect-substitution variant sets $(q, \epsilon_R, \epsilon_I) = (1, 0, 0)$. Reduced-price variants set $f = 0.01$ or 10^{-4} with instantaneous access; other parameters remain at reference settings. A material-speed variable is not allowed to make numerical prices exactly zero.

Corner classification uses the exact inequalities, not optimizer-selected shares. The stored optimal shares are checked separately using the objective and concavity certificates described in the archived Supplementary Material and audit outputs. Small platform-dependent changes

Table S2: All-inward inputs out of 1,024, with a conditional price-robustness check. The last column applies a uniform 10% price reduction to each listed scenario while retaining its other settings. It is computed from the exact boundary, not from a new behavioral assumption or an observational sample.

Scenario	All inward	Still inward after 10% price cut
Sublight material travel	929	923
Fast communication only	929	923
Light-speed limit	326	313
Travel and communication at 10c	234	219
Travel and communication at 1000c	219	207
Instantaneous reference	219	207
No experiential substitution	146	135
Large residual values	14	14
No substitution + large residuals	11	10
Local productivity $\times 10$	314	296
Local and remote productivity $\times 10$	222	211
Perfect substitution; no residual values	527	505
Outward price floor $\times 0.01$	0	0
Outward price floor $\times 0.0001$	0	0

in transformed parameters or nearly tied optimized shares do not change any archived corner classification in the checks reported here.

For design-measure sensitivity, Sobol seeds 20260911 through 20260918 each generate 2^{12} points. Both log-uniform and linear-uniform transformations use the same bounds in table S1. Repeated scrambles assess numerical integration stability under a selected measure. They do not supply a prior justified by extraterrestrial observations.

S3 Archived two-period construction

A one-horizon budget comparison omits one of the strongest reasons to expand: external industry can enlarge future resources. A minimal two-period construction makes this effect explicit while also allowing local investment to be productive. Let $B_1 = 10$, $z_1 = B_1 - p_1 x$, and define

$$C_1(x) = C_{01} + a_1 \ln(1 + z_1) + r_1 x,$$
$$B_2(x) = B_{20} + g_L z_1 + g_X x, \tag{83}$$
$$W(x) = \ln[1 + C_1(x)] + \beta V_2[B_2(x)], \quad 0 \le x \le B_1/p_1. \tag{84}$$

Here $B_1 > 0$, $B_{20}, C_{01}, C_{02} \ge 0$, $a_1, a_2, p_1, p_2, r_1, r_2 > 0$, and $0 \le \beta \le 1$. The parameters $g_L, g_X \ge 0$ convert current allocations to next-period resources. Both directions receive a future-production channel. All represented downstream benefits of first-period projects are captured through B_2; this is not a full persistent-capital or interstellar replication model.

The continuation value is the second-period pure-computation problem,

$$V_2(B) = \max_{\substack{z_2, x_2 \ge 0 \\ z_2 + p_2 x_2 \le B}} \ln[1 + C_{02} + a_2 \ln(1 + z_2) + r_2 x_2]. \tag{85}$$

For $r_2 > 0$,

$$z_2^*(B) = \min\{B, \max(a_2 p_2 / r_2 - 1, 0)\}, \qquad x_2^*(B) = \frac{B - z_2^*(B)}{p_2}. \tag{86}$$

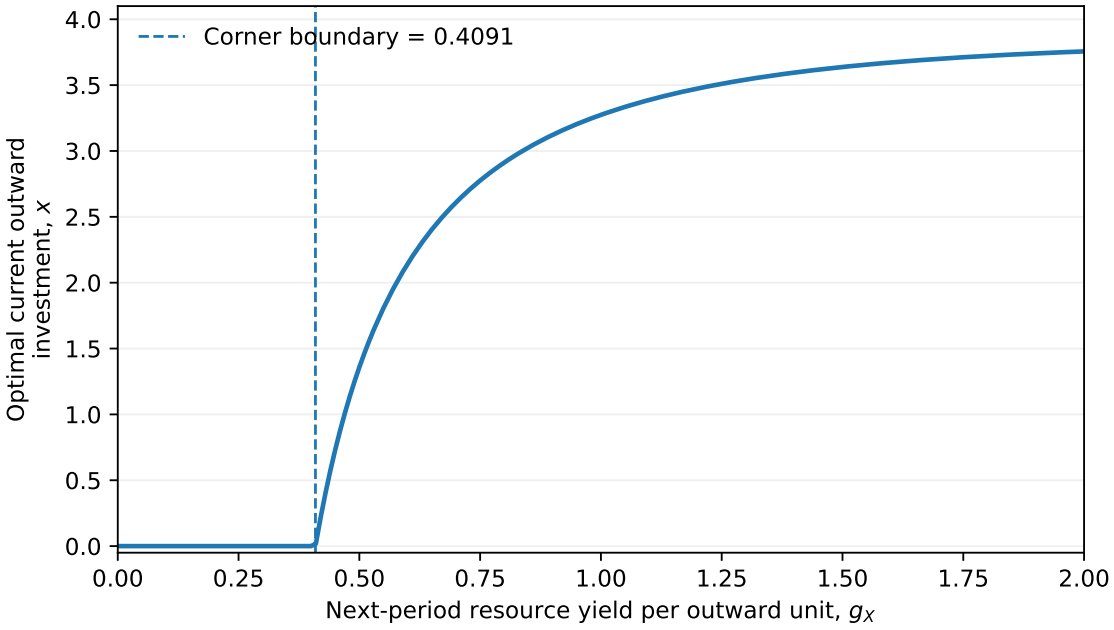


Figure S3: Two-period resource feedback. Local investment generates future resources at $g_L = 0.2$ per unit. Increasing the external yield g_X eventually overturns the all-inward present-only optimum. Parameters are normalized; the transition is an exact derivative boundary checked against a separate three-variable numerical optimization.

Its marginal value $V_2'(B)$ is the current best computational marginal return divided by $1+C_2^*(B)$: $a_2/(1+B)$ on the all-local branch, or r_2/p_2 on the mixed/external branch. The expressions agree at the switching budget. V_2 is increasing and concave.

The archived specialization of proposition 18 is

$$\frac{r_1 - p_1 a_1/(1+B_1)}{1 + C_{01} + a_1 \ln(1+B_1)} + \beta V_2'(B_{20} + g_L B_1)(g_X - p_1 g_L) \le 0. \tag{87}$$

The derivative proof in the main text applies because both current and continuation objectives here are concave.

The first term is the original within-period comparison. The second values the *difference* between external and local production of future resources. Counting g_X while setting g_L to zero without justification would bias the test against inward development; omitting g_X would bias it in favour. This extension is a structural check, not a calibrated decision rule for actual civilizations.

For $p_1 = p_2 = r_1 = r_2 = 1$, $a_1 = a_2 = 100$, $C_{01} = C_{02} = B_{20} = 1$, $g_L = 0.2$, and $\beta = 0.9$, present-only investment is all inward. The future-adjusted corner changes at $g_X = 0.4091$. At $g_X = 0.2$ both periods remain all inward. At $g_X = 2$, optimal first-period outward investment is $x = 3.7559$, leaving $z_1 = 6.2441$ locally and producing $B_2 = 9.7607$. All second-period investment is still local. Thus the optimal path can include *external investment undertaken to finance later inward activity*. This differs from claiming that external expansion is never useful. Figure S3 gives the full 201-setting construction.

The converse reversal is also possible. Keep $B_1 = 10$, $p_1 = p_2 = r_1 = r_2 = 1$ and $\beta = 0.9$, but set $a_1 = 5$, $a_2 = 100$, $C_{01} = C_{02} = 1$, $B_{20} = 0$, $g_L = 0.1$ and $g_X = 0$. The static term in equation (87) is $+0.03899$, and the present-only optimum is $x = 6$. The continuation term is -0.06310, making the total -0.02411. Concavity therefore makes $x = 0$ the two-period optimum. Local investment, rather than external investment, produces the more valuable continuation

budget in this example. Supplementary Material section S3 reports scalar and joint-optimization checks of both cases.

The two reversals are retained as finite-horizon examples. The infinite-horizon benchmark in proposition 16 now supplies a separate verification theorem, with its own production and growth assumptions. The nonlinear examples here are exact special cases of proposition 18, not of the linear benchmark.

S4 Additional identities and new verification files

The workload ratios used in the main text are, explicitly,

$$\frac{D_k(d+1)}{D_k(d)} = b + \frac{1}{W_d} \longrightarrow b, \tag{88}$$

$$F(C_{d+1}) - F(C_d) = \ln \frac{1 + c(bW_d + 1)}{1 + cW_d} \longrightarrow \ln b. \tag{89}$$

Their proofs divide by $W_d \to \infty$. A nonconcave example retained from v7.3 is

$$U(x) = \ln\{1 + 2\ln(2 - x) + 0.1x\} + 3x^2, \quad 0 \le x \le 1, \tag{90}$$

with $U'(0) \simeq -0.3772$, $U(0) \simeq 0.8697$, and $U(1) \simeq 3.0953$. It violates the concavity assumption, not the corner theorem.

Each new script in `code/validate_v8_*.py` is self-contained and prints explicit pass and fail counts. The accompanying README maps files to propositions and distinguishes assertions from independent randomized cases; each script records its seed and tolerances. The new no-replication classification reads `archive/v7_3/data/allocation/design.npz` without changing it. It solves observation on the restricted face, then evaluates the two replication reduced derivatives; 64 of the 1,024 cases are additionally compared with full constrained optimization. General restricted-face tests use 320 separate randomized instances with optional backups. Thus the full optimization check is sampled, whereas every frozen row receives the exact face criterion.

The original archive scripts have been rerun in a working copy, not in the immutable archive. Their outputs are included under `results/archive_rerun/`. No figures or archived classification counts have been revised. Mechanical source and PDF checks are recorded in `results/assembly_audit.json` and `results/final_audit.json`; physical-parameter sources are documented in the comments of `code/validate_v8_7_physical.py`. The finite-efficiency tests are in `code/validate_v8_8_ceiling.py` and `results/v8_8_ceiling.json`.